

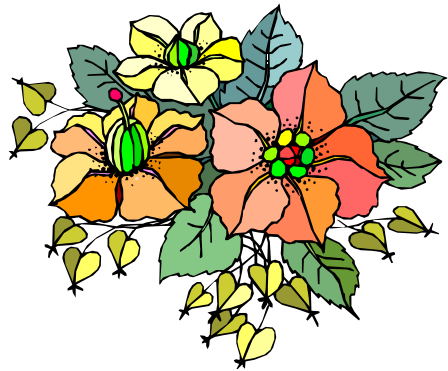
Darah, Organ Hematopoetik dan Sistem Retikuloendotelial



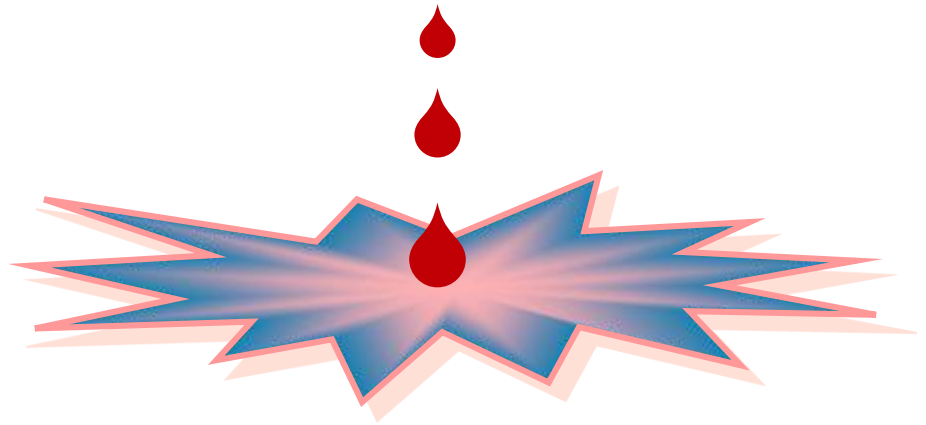
Dr. TRI AGUSTI SHOLIKAH, dr., M.Sc
Laboratorium Histologi FK UNS

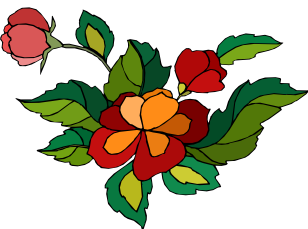
Outline

- Komponen darah : plasma dan sel darah
- Organ hematopoetik
- Organ Limfoid



KOMPONEN DARAH

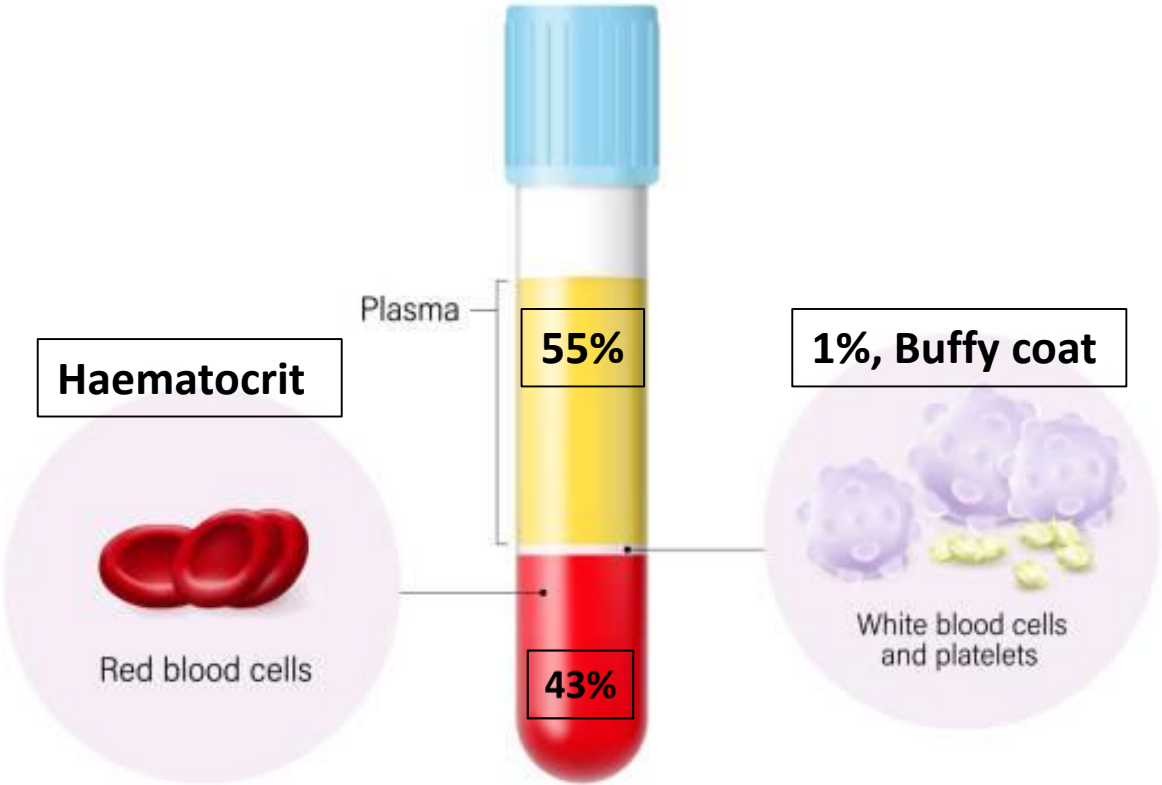




Darah + Antikoagulan



Sentrifuge



PLASMA



Plasma		
Water 92% by weight	Proteins 7% by weight Albumins 58% Globulins 37% Fibrinogen 4% Regulatory proteins 1%	Other solutes 1% by weight Electrolytes Nutrients Respiratory gases Waste products



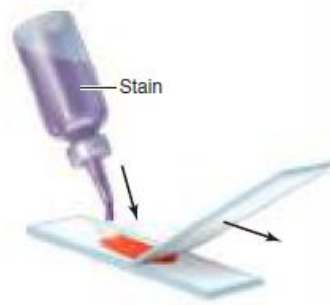
SEL-SEL DARAH



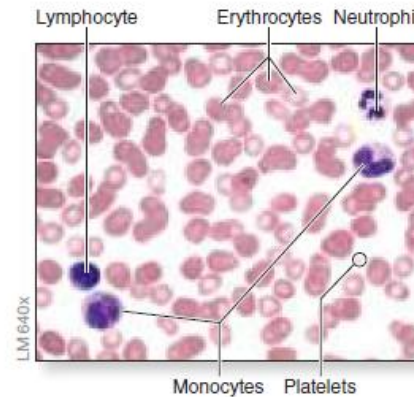
- ① Prick finger and collect a small amount of blood using a micropipette.



- ② Place a drop of blood on a slide.



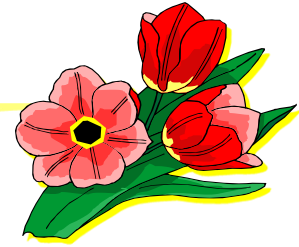
- ③a Using a second slide, pull the drop of blood across the first slide's surface, leaving a thin layer of blood on the slide.
- ③b After the blood dries, apply a stain briefly and rinse. Place a coverslip on top.



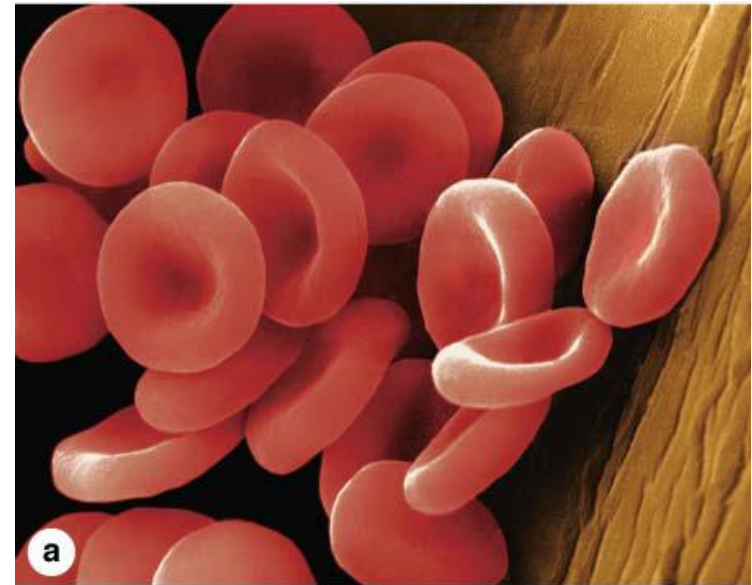
- ④ When viewed under the microscope, blood smear reveals the components of the formed elements.

- ✧ Eritrosit
- ✧ Leukosit
- ✧ Trombosit

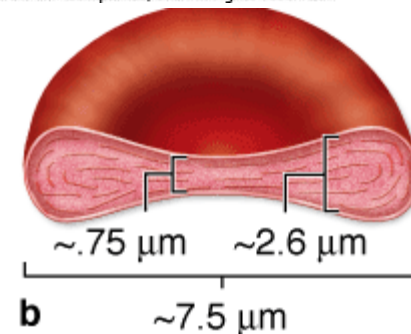
ERITROSIT (SEL DARAH MERAH)



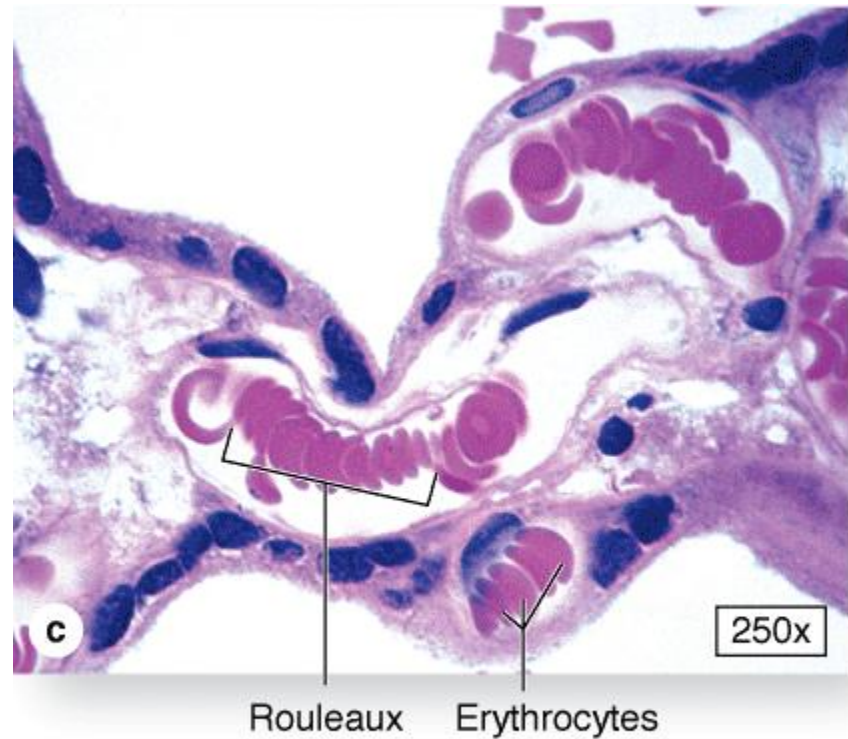
- * ♂ : $\pm 5,4$ juta / mm^3
- * ♀ : $\pm 4,8$ juta / mm^3
- * Sferis
- * Bikonkaf
- * Tidak mempunyai inti & Organela
- * Diameter $\pm 7,5 \mu\text{m}$
- * tebal tepi $\pm 2,6 \mu\text{m}$
- * tebal tengah $\pm 0,8 \mu\text{m}$.



Source: Mescher AL: *Junqueira's Basic Histology: Text and Atlas, 12th Edition*: <http://www.accessmedicine.com>
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Rouleaux Formation



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**PROTEIN
ERITROSIT**



**Membran plasma: ankyrin, spectrin, protein band 3 →
sel / membrane flexibel**

**Sitoplasma: Haemoglobin → mengikat dan
melepaskan O₂ atau CO₂ (proses respirasi)**



KEADAAN ABNORMAL ERITROSIT

- ◆ **Anemia**
- ◆ **Polycythemia / erythrocytosis**
- ◆ **Makrosit : diameter $> 9 \mu\text{m}$.**
- ◆ **Mikrosit : diameter $< 6 \mu\text{m}$.**
- ◆ **Anisositosis : ukuran bervariasi.**
- ◆ **Poikilositosis : bentuk abnormal.**

LEUKOSIT (SEL DARAH PUTIH)

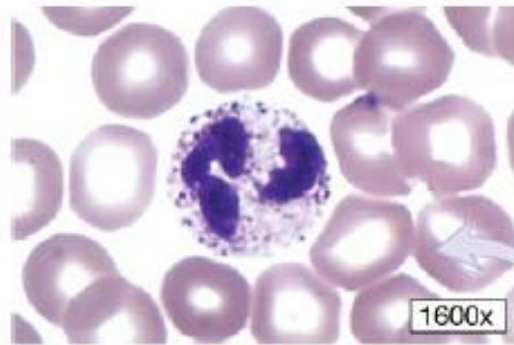
Berdasarkan ada tdk-nya granula spesifik
dan bentuk inti

Polimorfonuklear
granulosit

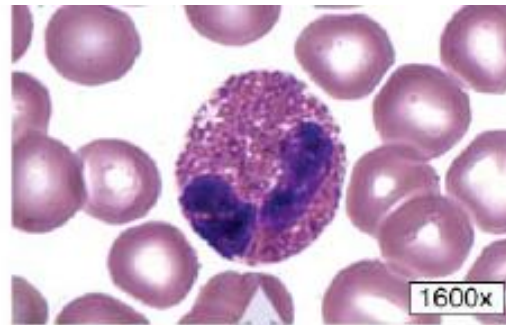


Mononuklear
agranulosit

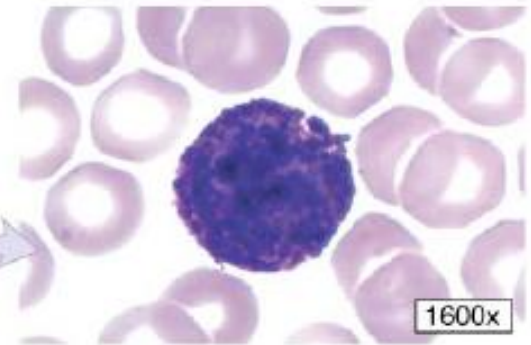
berdsrkan afinitas thd
zat warna asam/basa



Neutrophil



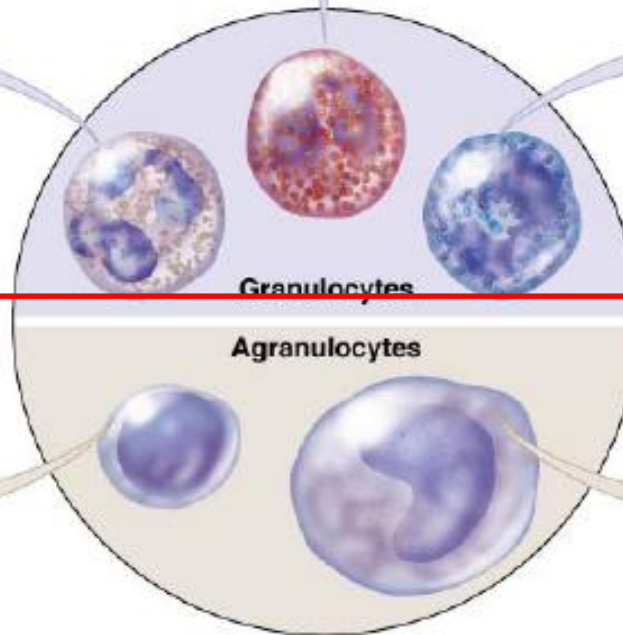
Eosinophil



Basophil



Lymphocyte



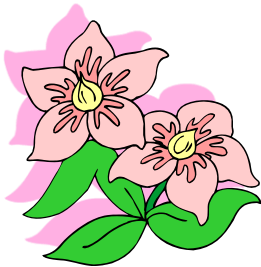
Monocyte



***Normal : 6000 – 10.000 / mm³**

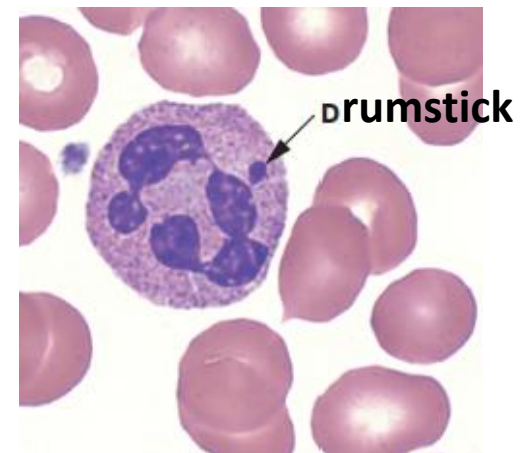
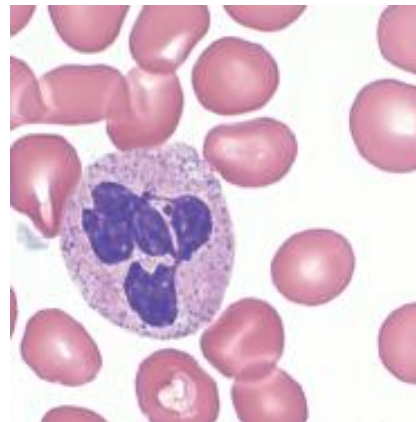
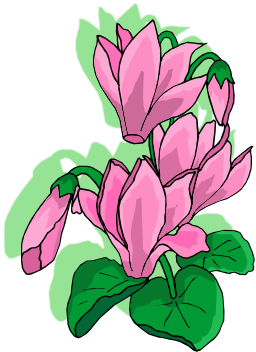
**jenis leukosit dlm persen ⇒
hemogram (hitung jenis leukosit) :**

- ♦ **Neutrofil : 60 – 70 %**
- ♦ **Eosinofil : 2 – 4 %**
- ♦ **Basofil : 0 – 1 %**
- ♦ **Limfosit : 20 – 30 %**
- ♦ **Monosit : 3 – 8 %**

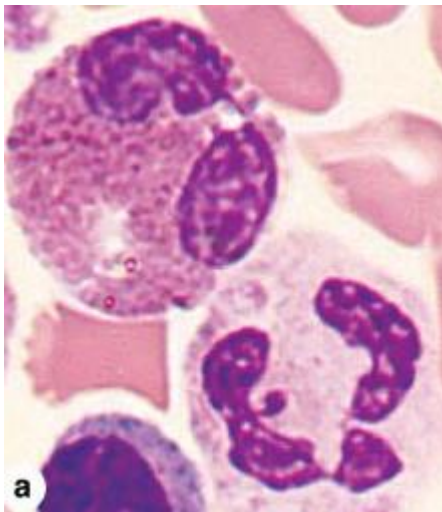
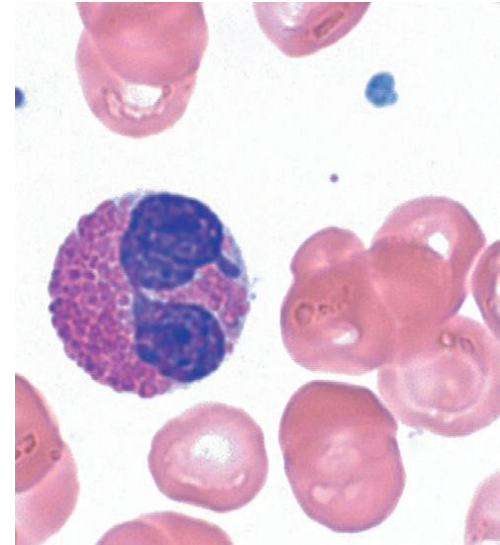


NEUTROFIL (PMN)

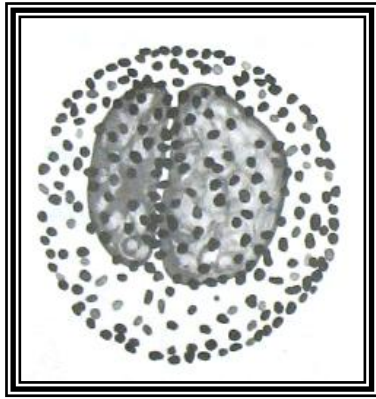
- Diameter : 12 – 15 μm
- Nukleus : 2-5 lobus
- ♀ : drumstick
- Granula spesifik : sangat kecil, hampir tdk tampak
- Fungsi : Fagositosis (terutama bakteri)



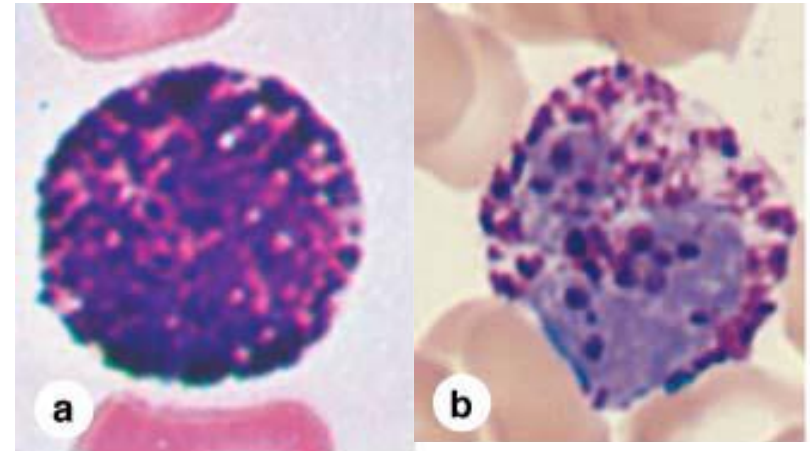
EOSINOFIL



- Diameter : 12 -15 μm .
- Nukleus : bilobus
- Granula spesifik : besar, merah (dicat eosin)
- Fungsi : Berperan dlm rx alergi & infeksi parasit (*major basic protein*). Fagositosis lbh lambat & lbh selektif dp neutrofil.



- sulit ditemukan pd *blood smear*.
- Diameter : 12 – 15 μm .
- Nukleus : 2/> lobus ireguler
- Granula spesifik : besar, menutupi nukleus
- Fungsi: mengeluarkan mediator inflamasi pd rx alergi (heparin dan histamin). Fagositosis sangat terbatas.

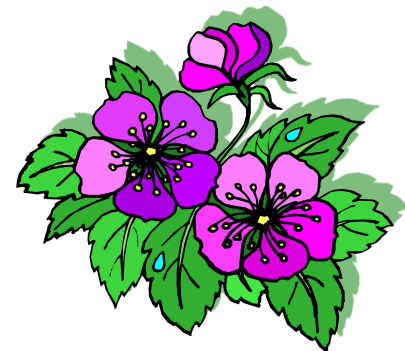


Source: Mescher AL: *Junqueira's Basic Histology: Text and Atlas*, 12th Edition: <http://www.accessmedicine.com>
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LIMFOSIT



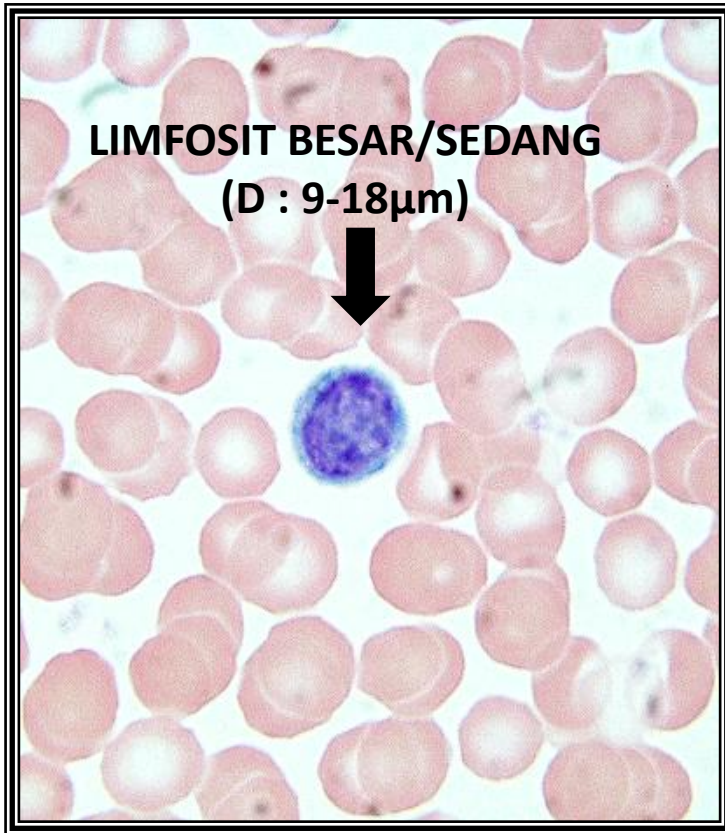
- **Nukleus** : sferis, kadang2 melekok
- **Sitoplasma** : sgt sedikit, lingkaran tipis mengelilingi nukleus
- **Berdsr Diameternya** : kecil (dominan), sedang dan besar
- **Fungsi** : Mekanisme kekebalan tubuh



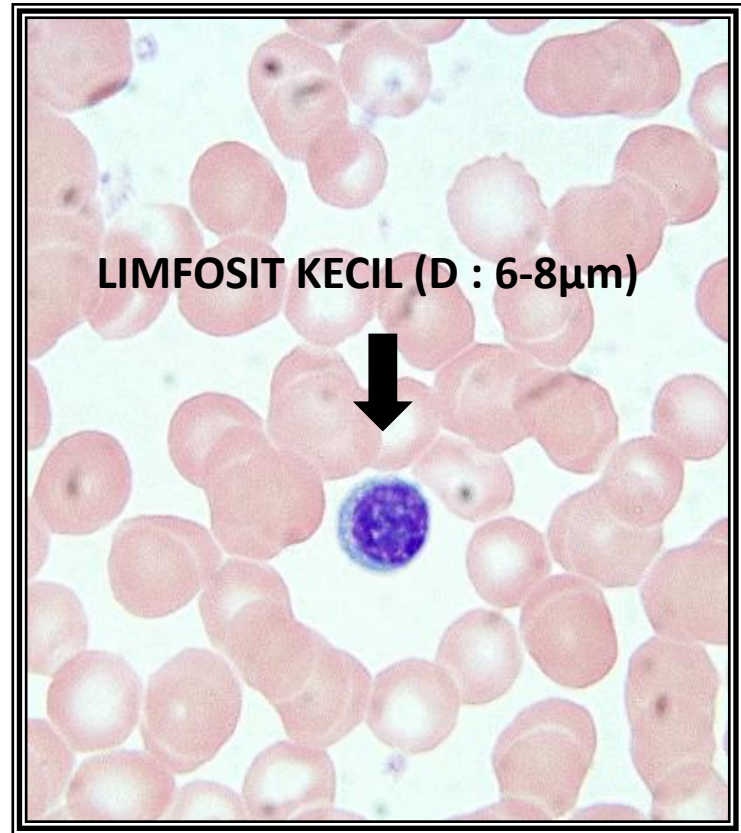


LIMFOSIT

LIMFOSIT BESAR/SEDANG
(D : 9-18 μ m)

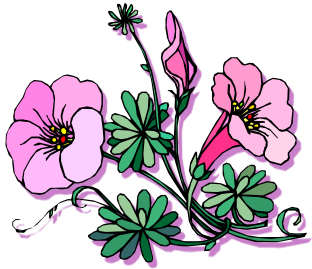


LIMFOSIT KECIL (D : 6-8 μ m)



KLASIFIKASI LIMFOSIT

- Brdsrkan reseptor pd permukaan sel : Limfosit T dan B → tdk bs dibedakan dg mikroskop
- Subpopulasi Limfosit T : sel Th (CD4), sel Tc (CD8)



Antigen masuk tubuh



Bertemu limfosit



Limfosit teraktivasi :

Ukuran > besar (limfoblas) Proliferasi & diferensiasi



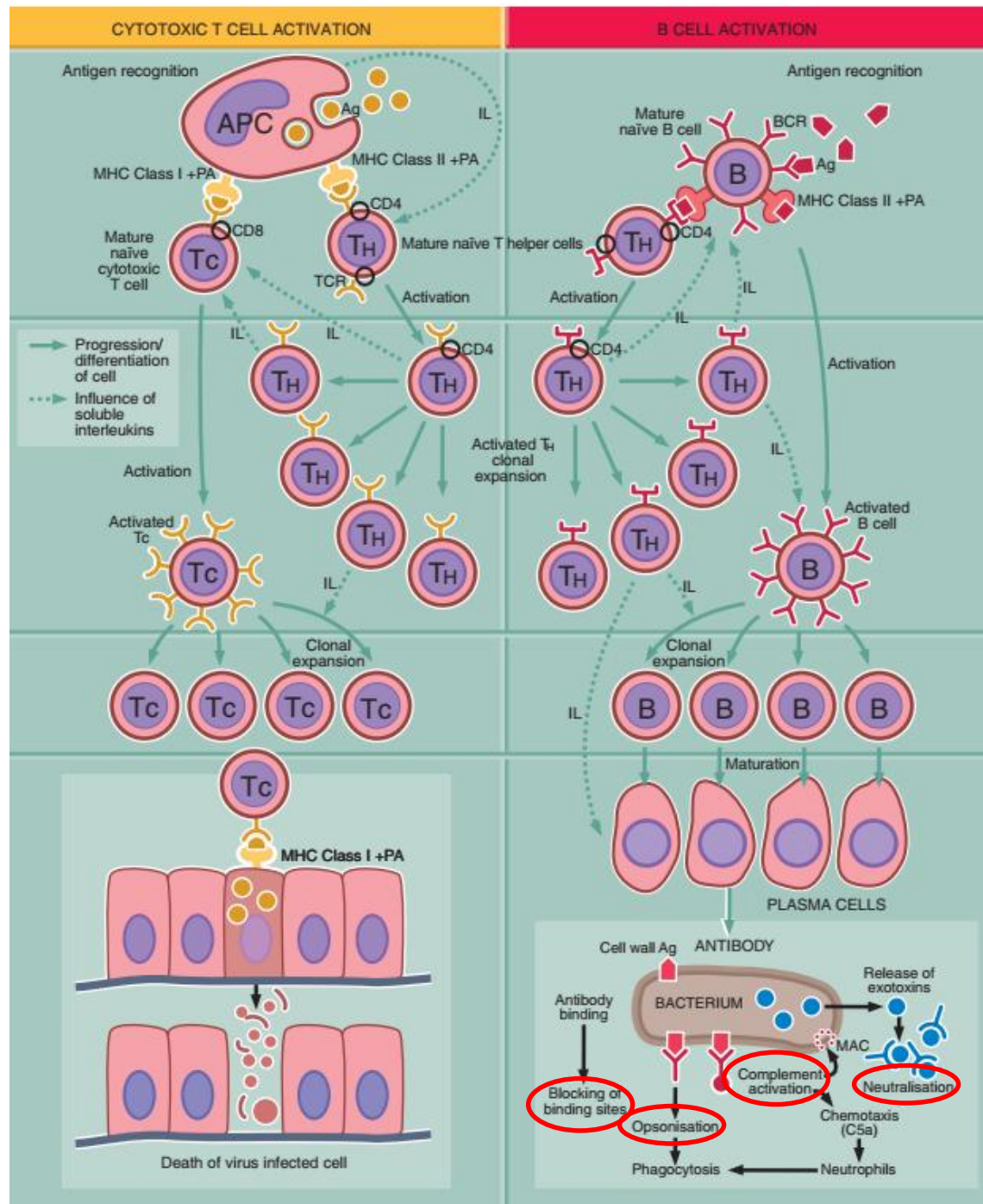
Sel efektor

Langsung
menghancurkan
antigen yang masuk

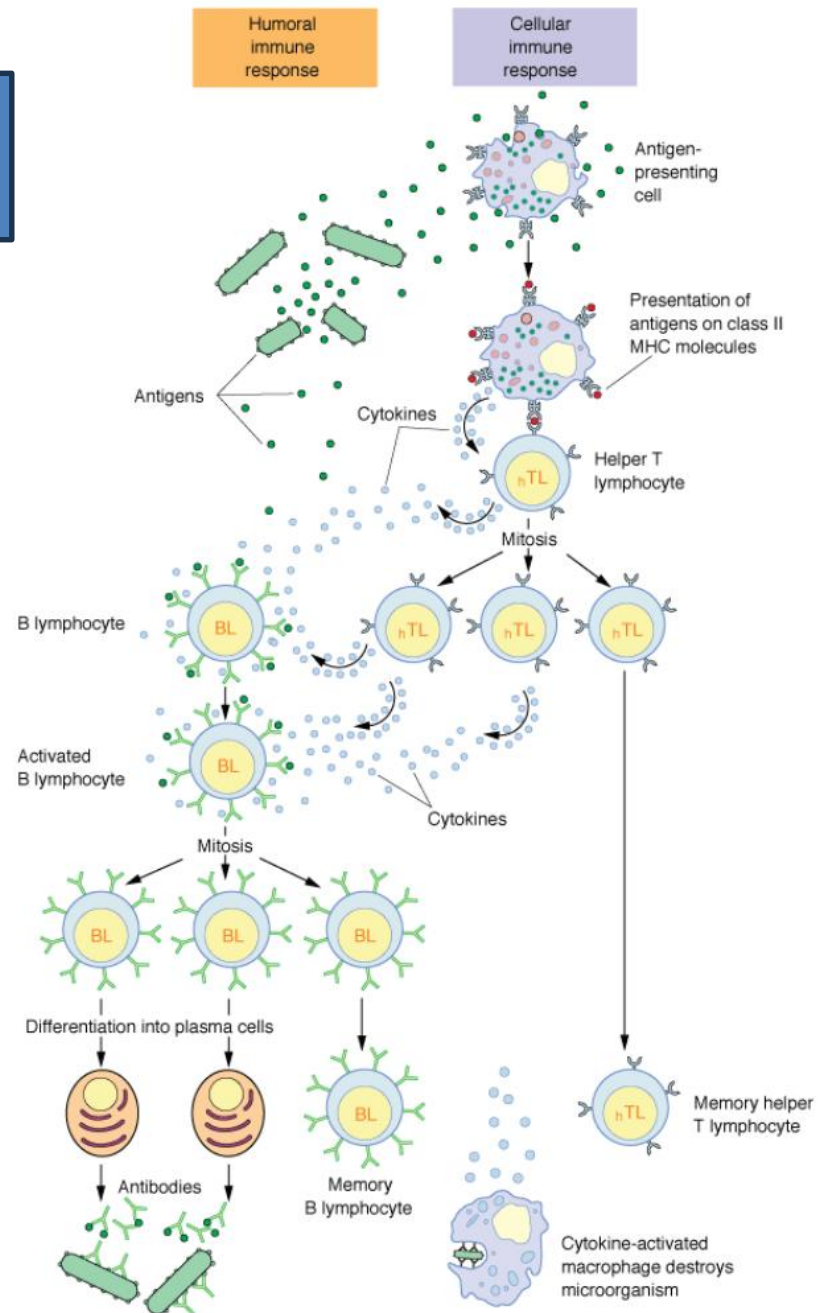
Sel memori

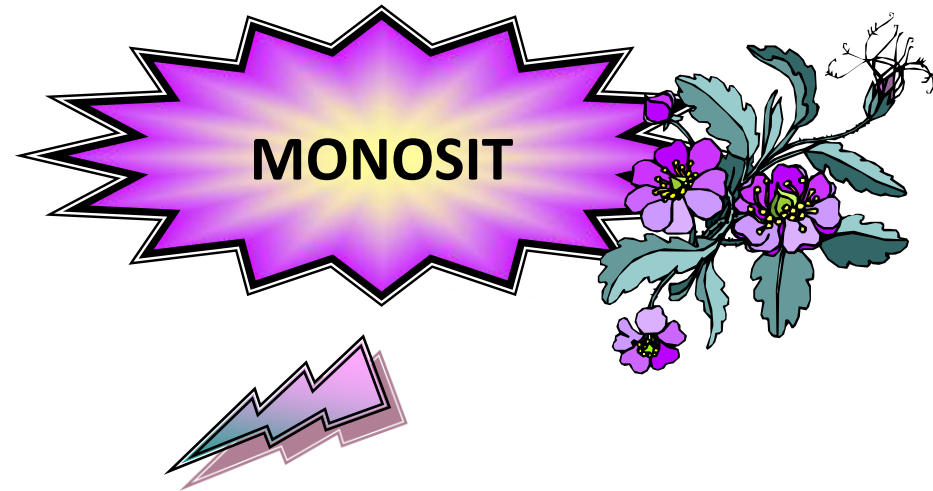
Sel pasif, mempunyai kemampuan yg
lbh cepat & kuat dlm menghancurkan
antigen bila bertemu dg antigen yg
sama di kemudian hari.

AKTIVASI
LIMFOSIT

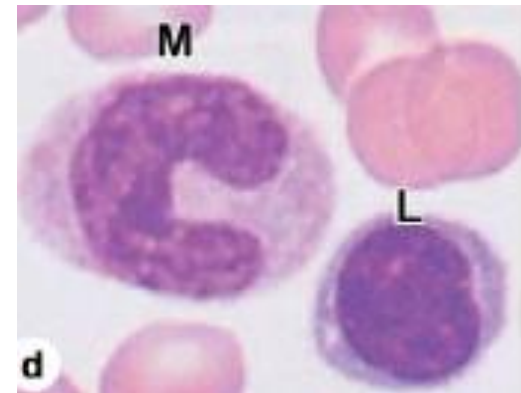
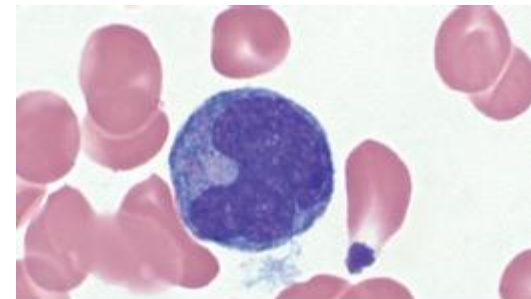


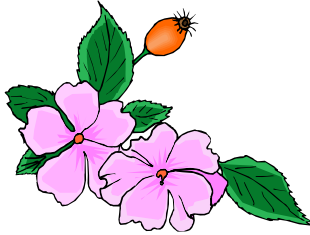
TYPES OF IMMUNE RESPONSES





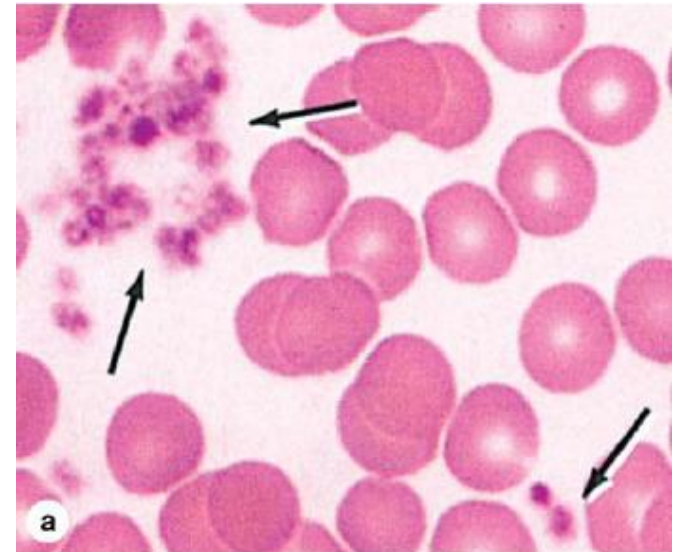
- Diameter : 12 – 20 μm .
- Mirip dg limfosit besar.
- Nukleus : besar, btk oval/spt ginjal / huruf U
- Fungsi : prekursor sel fagosit mononuklear $\rightarrow$ makrofag (jar.ikat), mikroglia (CNS), osteoklas (tulang)

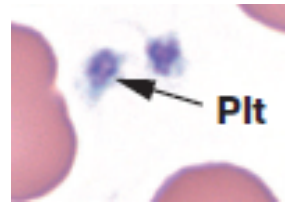
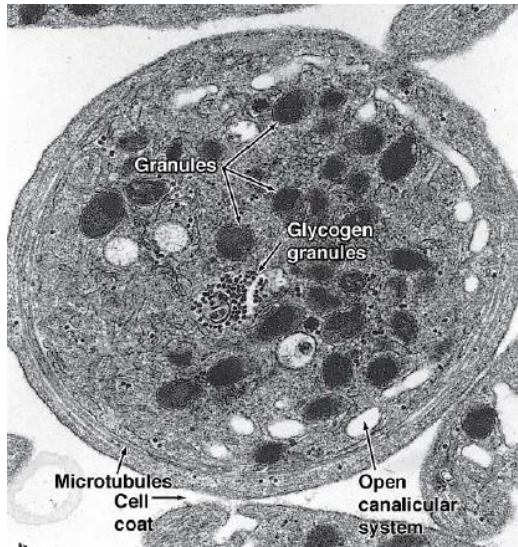




TROMBOSIT (KEPING-KEPING DARAH)

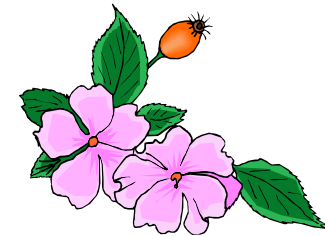
- Kadar Normal : 200.000–400.000/mm³.
- Hasil fragmentasi megakariosit.
- Bulat, diameter 2-4 μ m, bikonveks, inti (-).
- Pd *blood smear* : tampak agregasi





TROMBOSIT
• (KEPING-KEPING DARAH)

Pengecatan Romanowsky



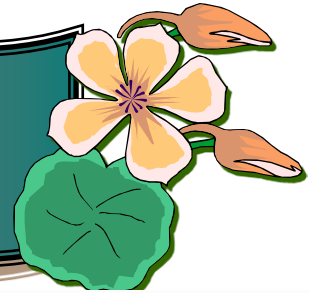
Granulomer/chromatomer/Chromomere

- Di bagian tengah
- Lebih tebal
- Membias sinar lebih kuat
- Terdpt granula alfa (protein pembekuan darah), delta (ADP, ATP & serotonin) & lambda (enzim lisosom); mitokondria serta glikogen.

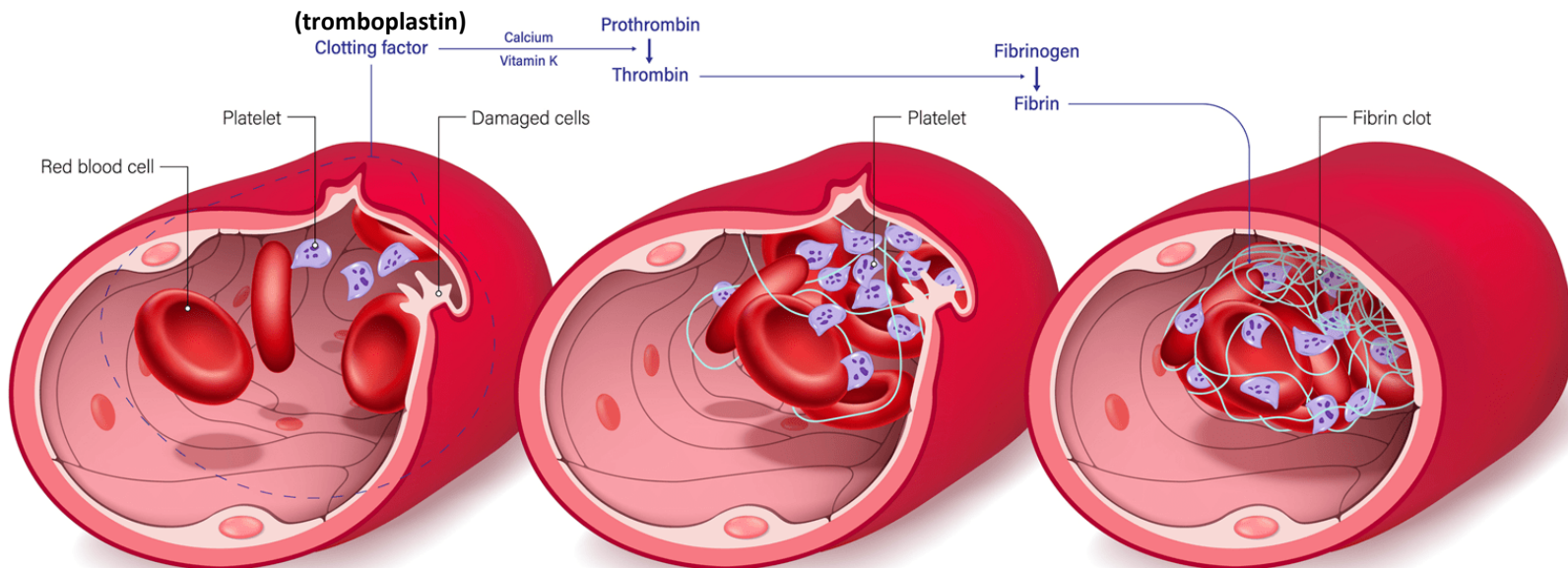
Hialomer

- Di bagian tepi
- Lebih tipis
- Homogen (biru pucat)
- Terdpt filamen untuk : mempertahankan btk trombosit, proses retraksi bekuan drh dan pembentukan pseudopodia.

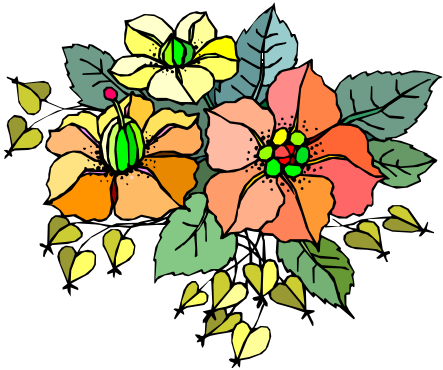
Fungsi trombosit pd proses pembekuan darah

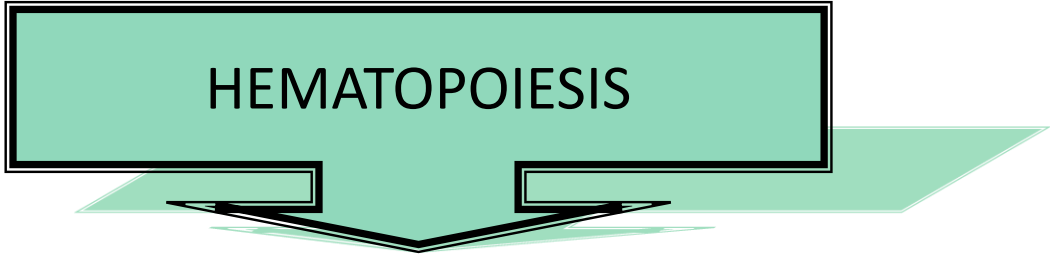


Blood Clotting



ORGAN HEMATOPOEITIK





HEMATOPOIESIS

Proses pembentukan dan perkembangan sel darah
Terjadi di organ hematopoietik
Eritropoiesis, leukopoiesis, trombopoiesis

HEMATOPOIESIS

- ✱ Prenatal (**yolk sac**; **hepar, lien**; **sumsum tulang**)
- ✱ Postnatal (**sumsum tulang**; nodus limfatikus, lien, timus; hepar)

Fisiologis:

Sumsum tulang: tempat pembentukan awal sel-sel darah

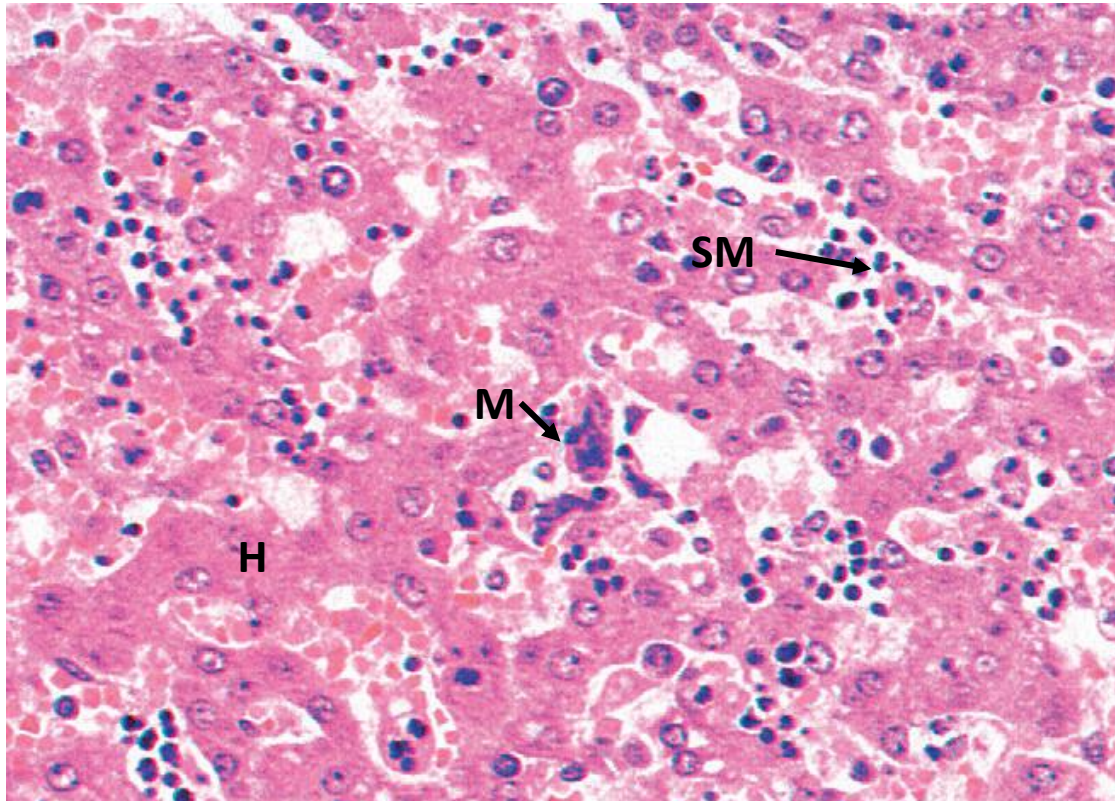
Nodus limfatikus, lien, timus: tempat perkembangan lbh lanjut (limfosit)

Patologis: nodus limfatikus, lien, timus, hepar dpt sbg tempat pembentukan sel-sel darah spt sumsum tulang

LIMFOPOESIS

- Limfosit B → produksi dan maturasi di sumsum tulang
- Limfosit T → produksi di sumsum tulang, maturasi di Thymus

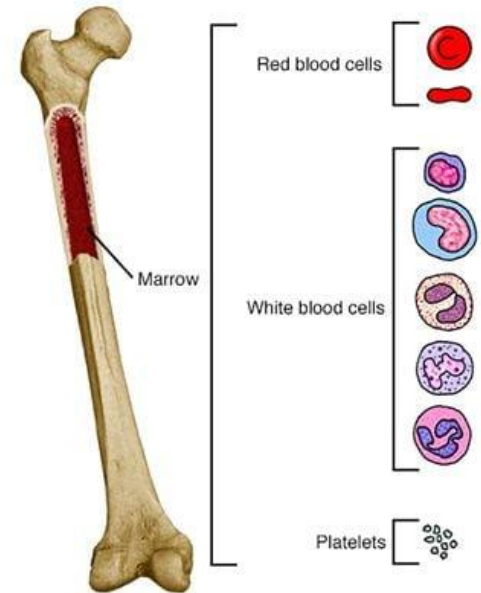
HEMATOPOIESIS PRENATAL (DI HEPAR)



SM= Sel hematopoetik
M= Megakariosit
H= Hepatosit

HEMATOPOIESIS POSTNATAL

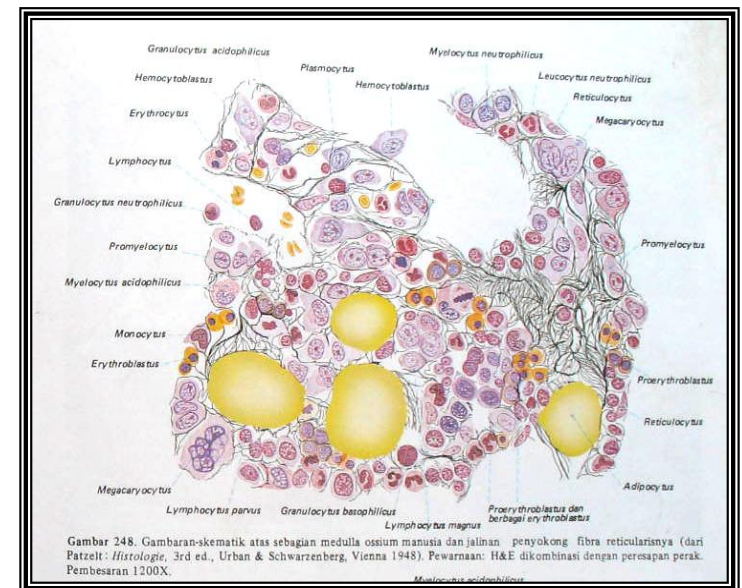
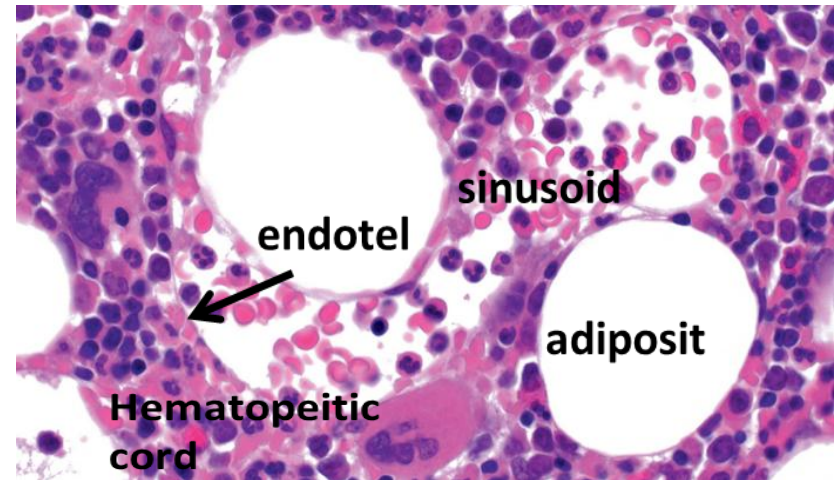
- * Sumsum tulang terdapat dalam saluran medula tulang panjang, tulang pipih, dan dalam rongga tulang spons.
- * Tipe sumsum tulang :
 - ✓ Sumsum tulang merah/hematogenus → tempat hematopoiesis
 - ✓ Sumsum tulang kuning → sel adiposa lebih banyak, Resting bone marrow (aktivitas haematopoiesis sedikit)



KOMPONEN SUMSUM TULANG

Sumsum tulang merah :

- **Stroma** ⇒ anyaman serabut retikuler dan sel-sel retikuler.
- **Hematopoietic cords** ⇒ Sel darah & prekursoranya (seri eritrosit, trombosit, granulosit, monosit, limfosit).
- **Sinusoid** (dilapisi sel endotel), arteri dan vena kecil
- Makrofag, sel adiposa.
- Matriks ⇒ kolagen tipe I dan III, proteoglikan, laminin, fibronectin, hemonektin.

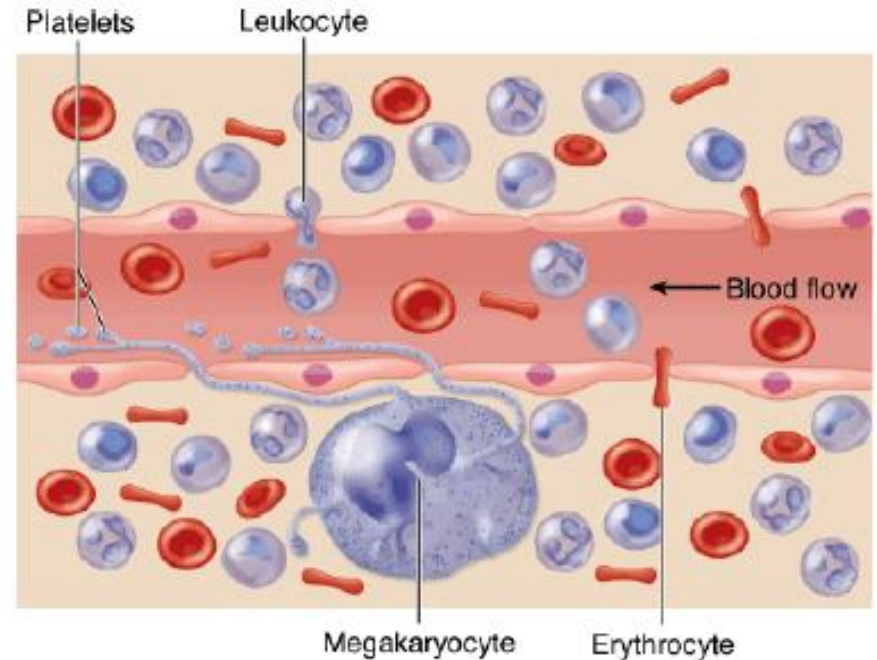


erythrocytes, leukocytes, and platelets
di sumsum tlg → sinusoid (mll celah
antar sel endotel) → sirkulasi.

erythrocytes → by a pressure gradient
across its wall.

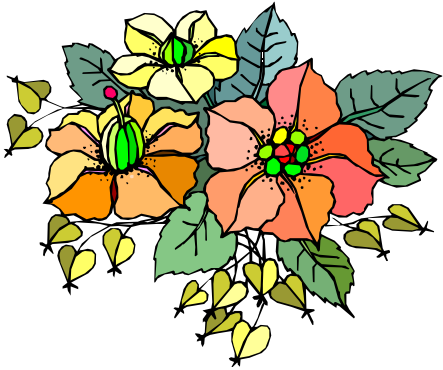
Leukocytes → by their own activity.

Megakaryocytes form thin processes
(proplatelets) that also pass through
such apertures and liberate platelets at
their tips.

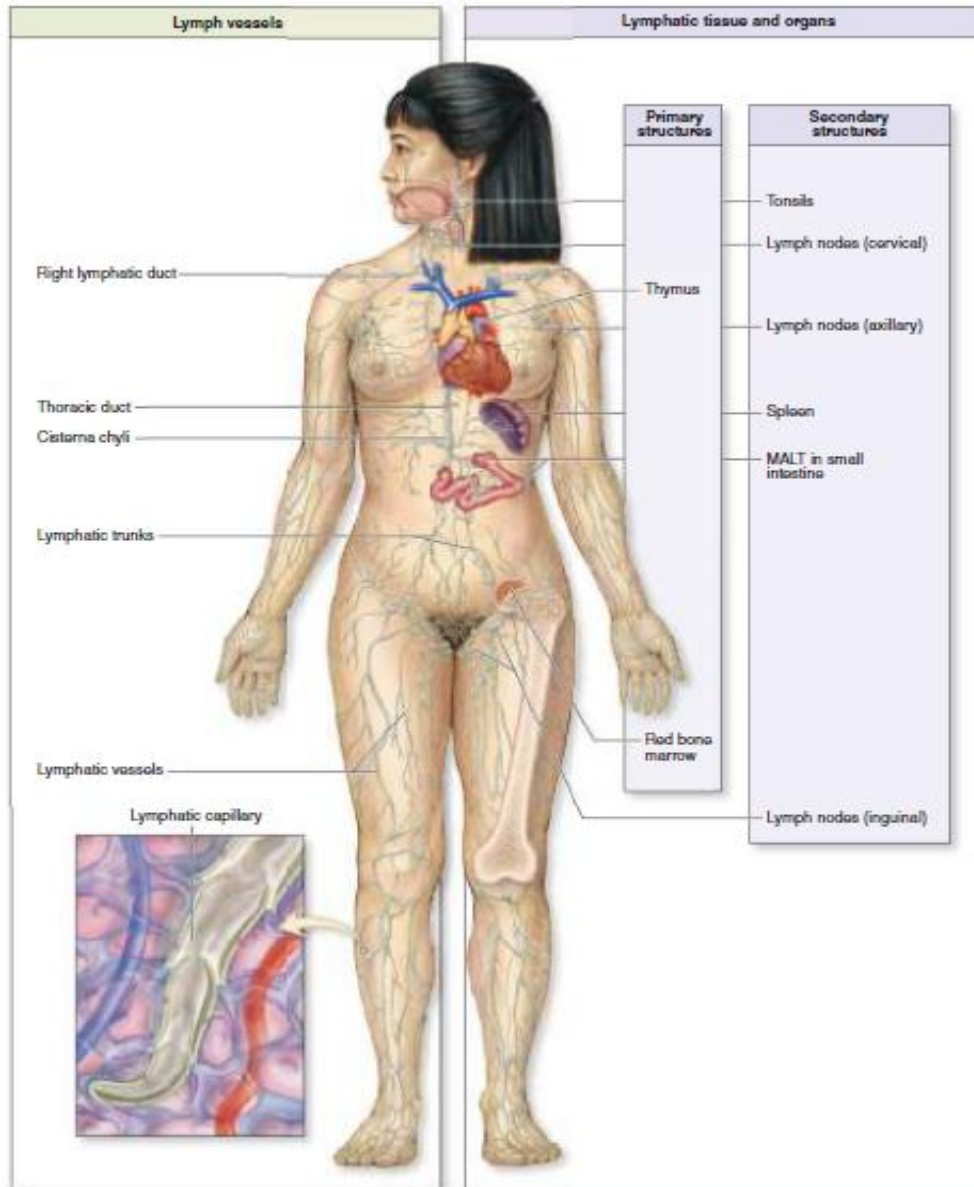


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ORGAN LIMFOID



Organ / jaringan limfoid

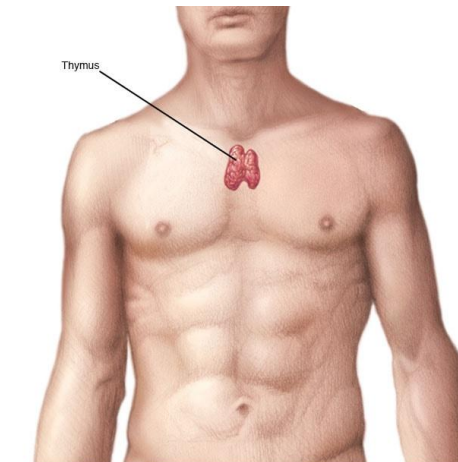
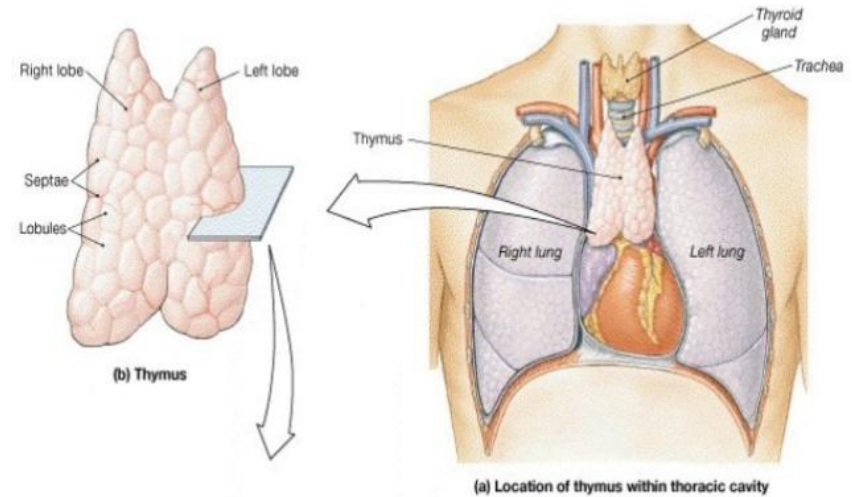


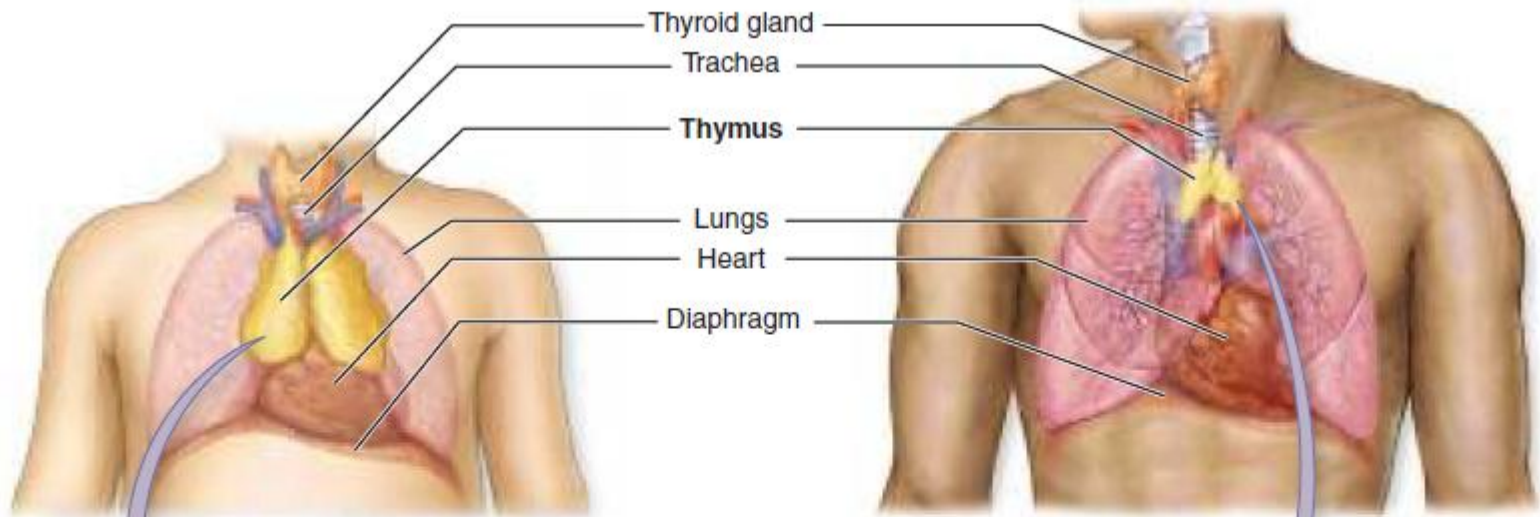
Klasifikasi:

1. Primary lymphoid organs
→ produksi dan maturasi limfosit
2. Secondary lymphoid organs (lymphocytes are activated in response to antigen)

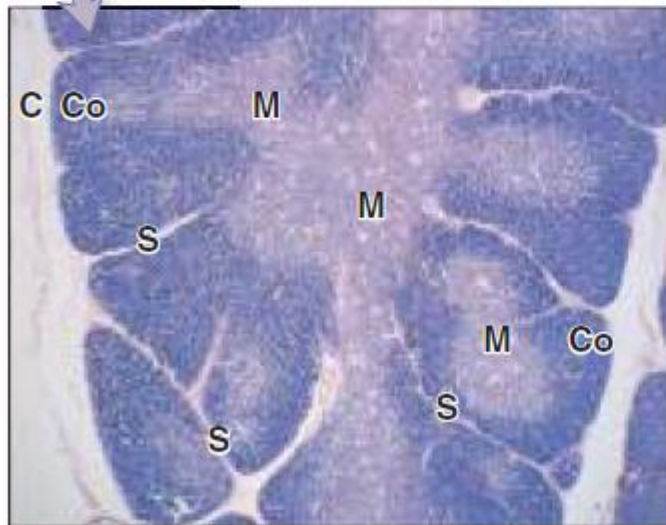
1. Thymus

- Lokasi: mediastinum anterior bgn atas, dibawah leher
- Berat saat pubertas : 30-40 g
- Sangat aktif pada masa anak-anak
- Saat dewasa: involusi (fatty infiltration and lymphocyte depletion)

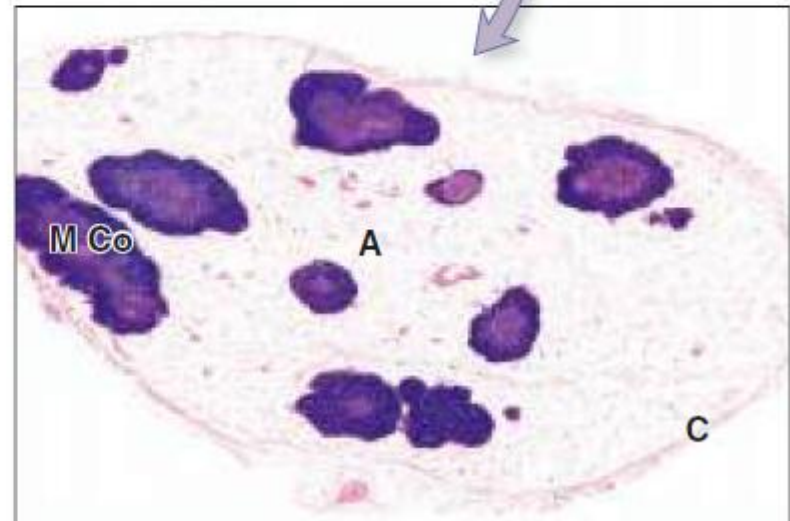




(a) Child (left) and adult (right) thorax, anterior view



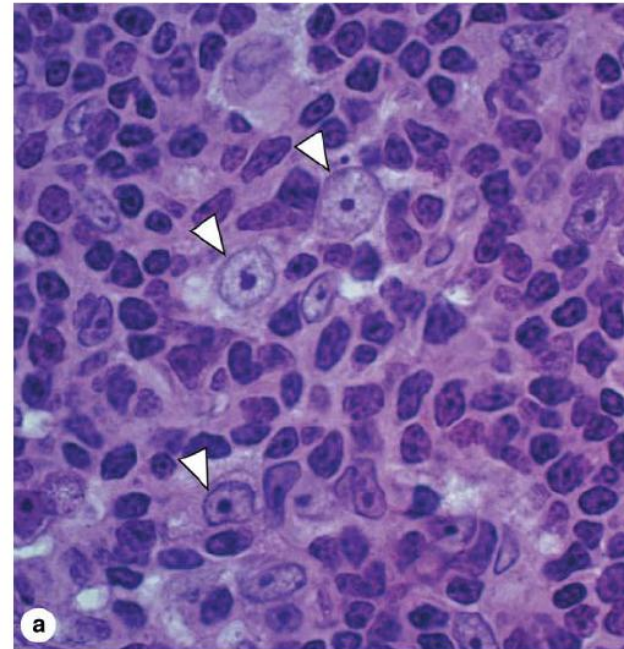
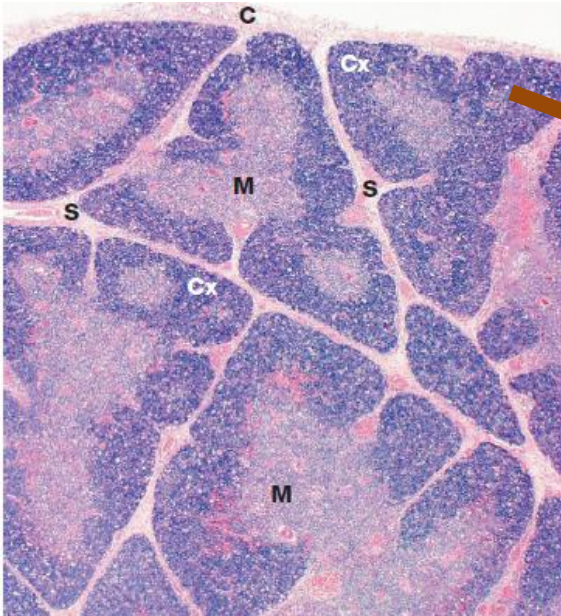
(b) Micrograph of child's thymus



(c) Micrograph of adult's thymus

Fatty infiltration of the interlobular septa occurs first, spreading out into the cortex and later the medulla

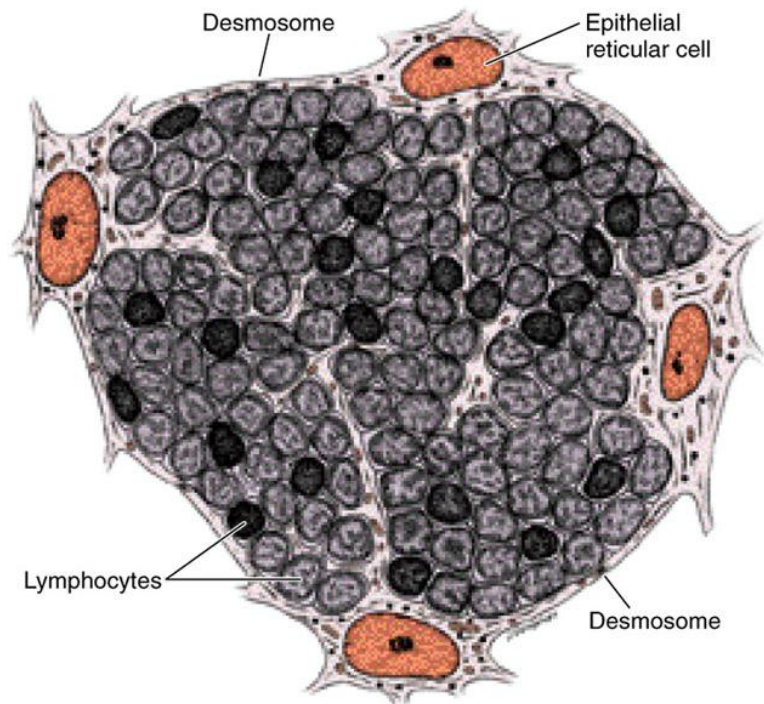
CORTEX THYMUS



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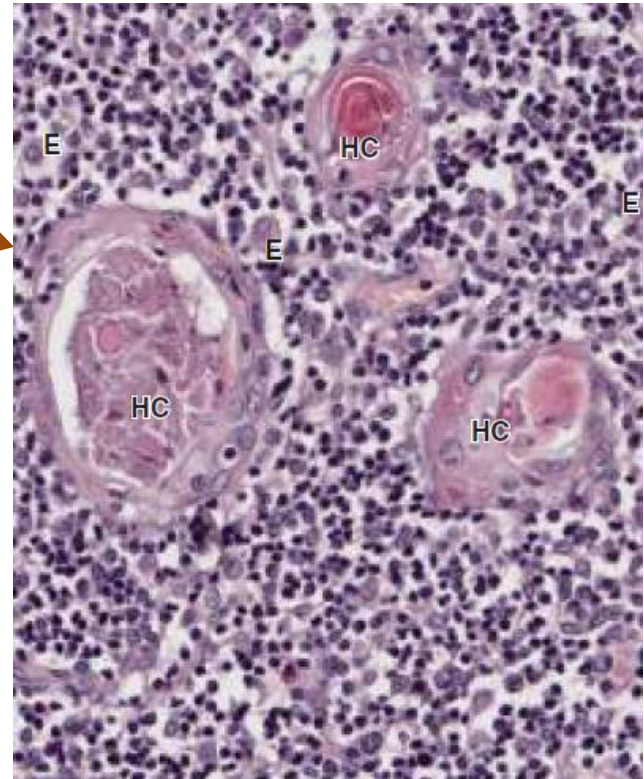
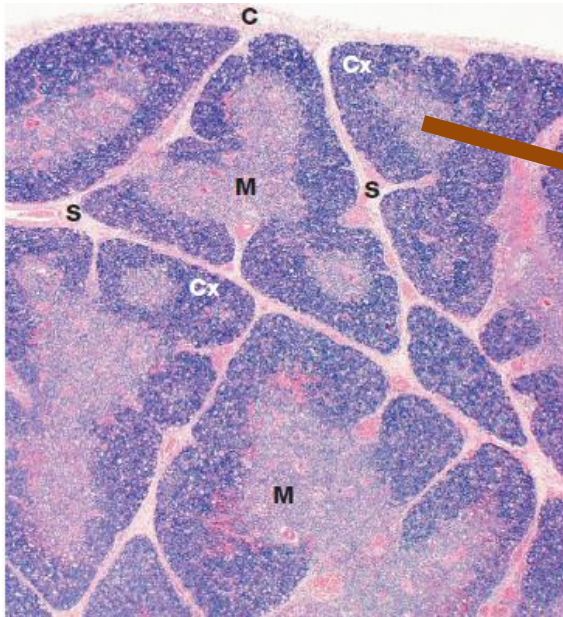
1. Stroma : jaringan epithelial reticular
2. Parenkim : limfosit T yg berukuran kecil (Thymosit)

Sel Epithel Retikuler



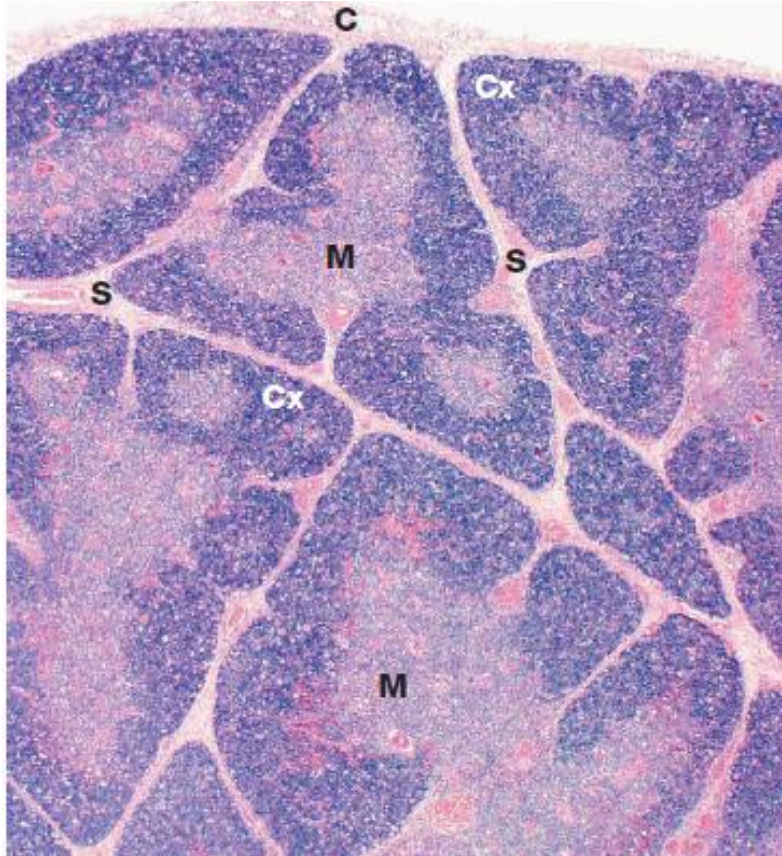
- Bentuk stelate
- Prosesus panjang
- Antar prosesus sel dengan sel lain dihubungkan desmosom
- Berperan sebagai sawar darah - tymus

MEDULLA THYMUS



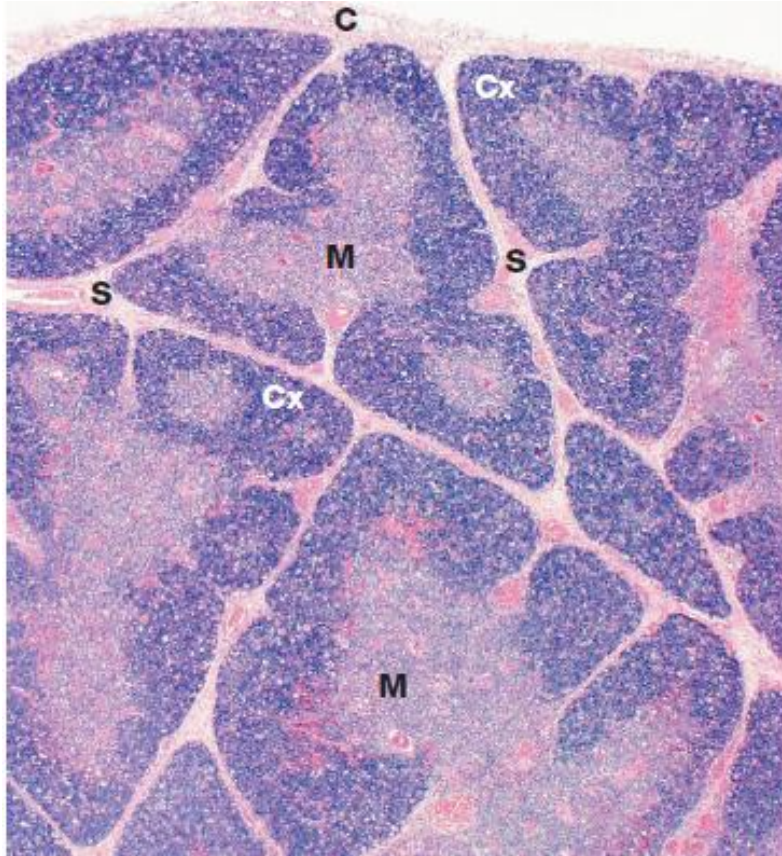
- Warna lbh pucat
- Corpusculum hasal (H): sel epitel retikuler tersusun konsentris, mgd keratin filamen dan terkalsifikasi (mrp btk degenerasi)
- Limfosit sedang dan besar
- Stroma : epitel retikuler

Vaskularisasi Thymus



- Arteri menembus kapsula → mengikuti jaringan ikat → masuk organ → septum sebagai arteriole → batas korteks-medula → kapiler → parenkim → venule
- Di korteks :
 - ✓ Arterioles and capillaries are sheathed by flattened epithelial reticular cells with tight junctions.
 - ✓ endotel kapiler : continuous & has a thick basal lamina → **blood-thymus barrier** → **melindungi thymosit dari Antigen.**

Vaskularisasi Thymus

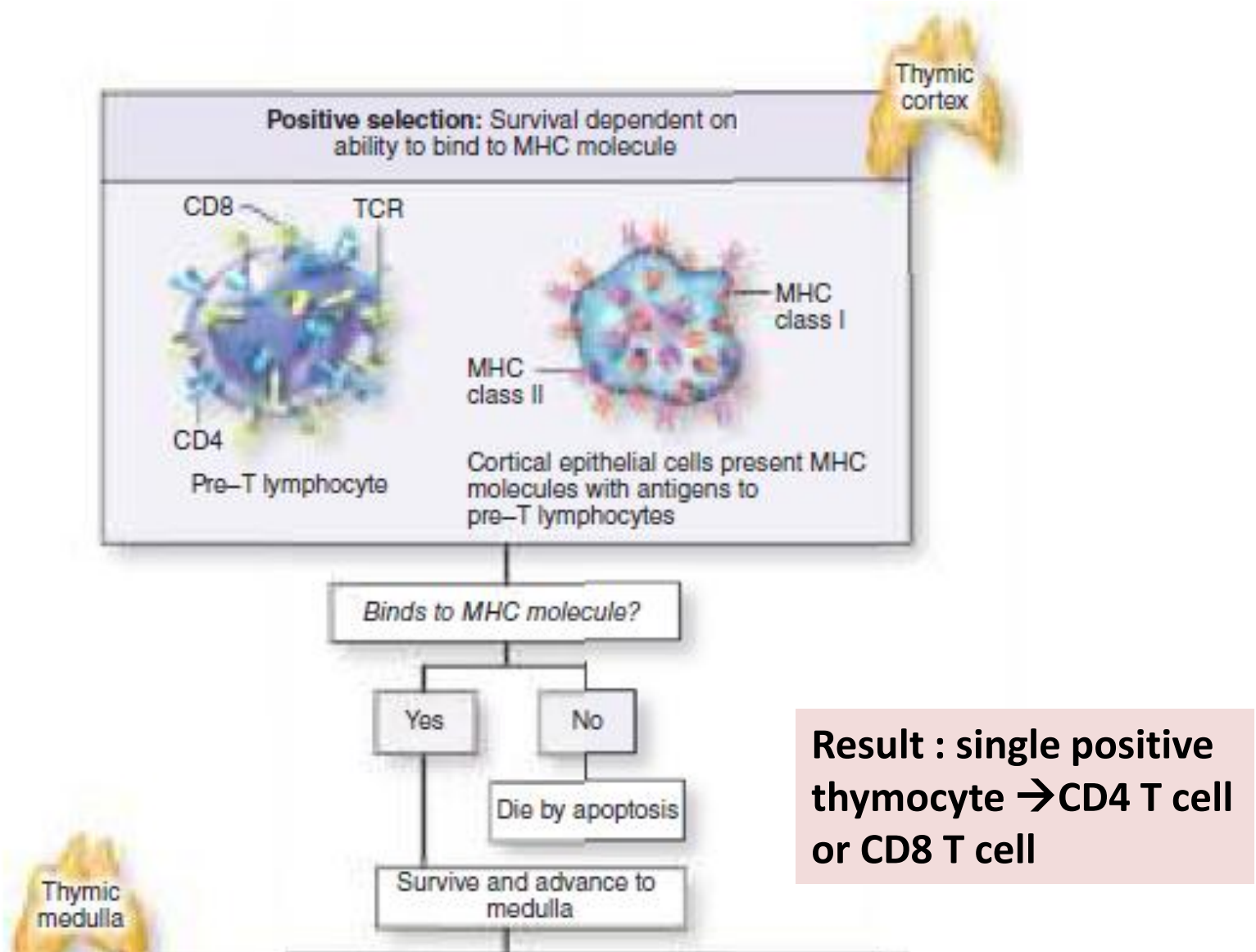


- Di medulla :
 - ✓ tdk ada barrier
 - ✓ mature T lymphocytes exit the thymus via venules in this zone.
- **no afferent lymphatic vessels** and does not constitute a lymph filter
- Semua pembuluh limfe (capsule, septa and blood vessel walls) → **efferent.**

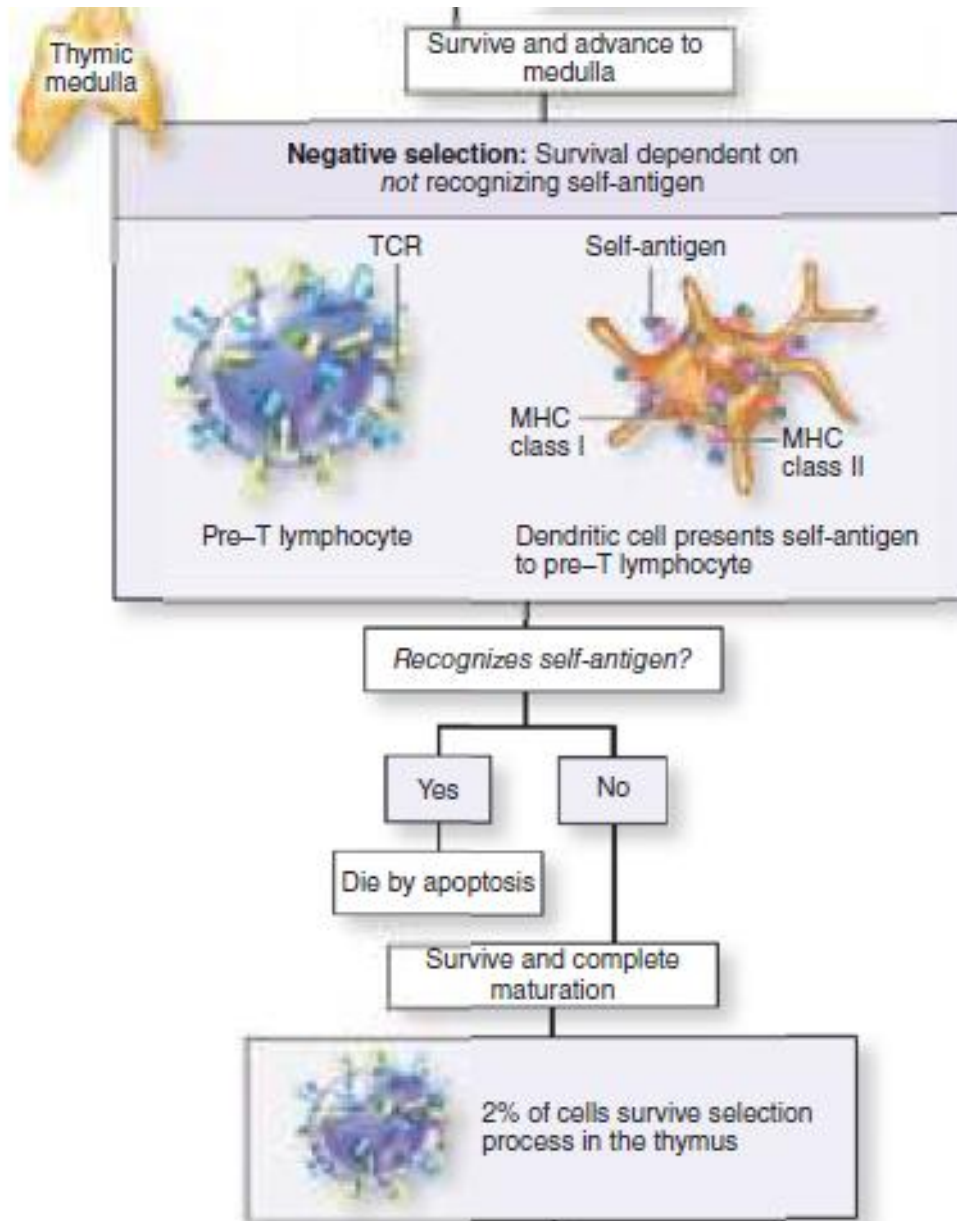
Fungsi Thymus

- Tempat maturasi limfosit T.
- Proliferasi limfosit T matur.
- Development of immunological self-tolerance → self-reactive T cells are removed, preventing the development of *autoimmunity*.
- Produksi hormon **thymopoietin, thymosins, thymulin & Thymus humoral factor** oleh sel apithel retikuler → mengatur maturasi, proliferasi dan fungsi sel T di thymus dan peripheral lymphoid tissues. .

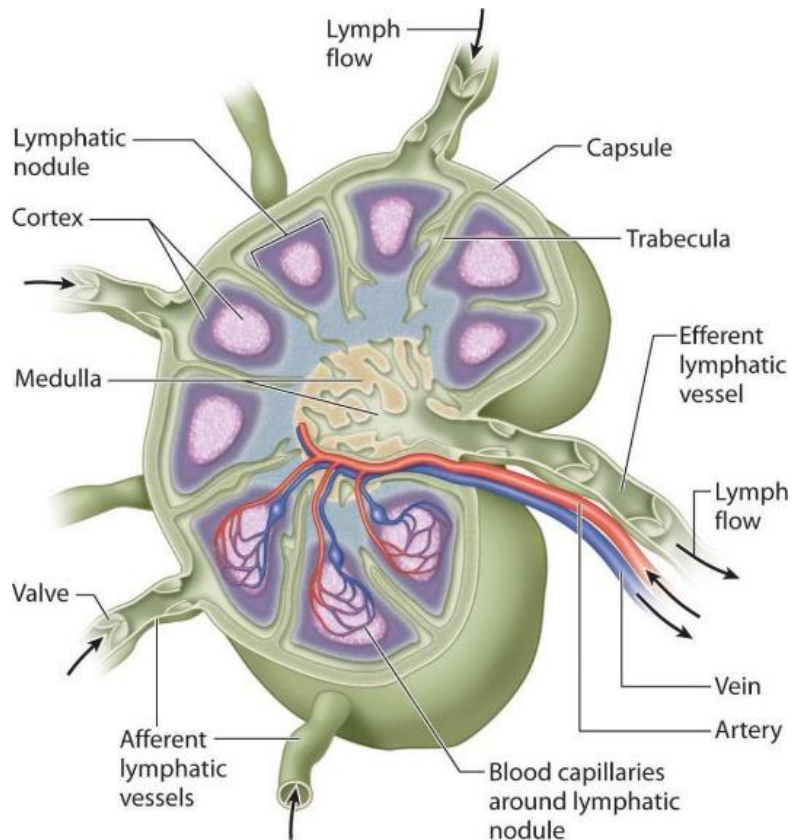
Maturasi Limfosit T di Thymus



Maturasi Limfosit T di Thymus



2. LIMFONODUS



- Bentuk seperti ginjal, berkapsul, $\varnothing$: 2-10 mm
- Terdapat di sepanjang pembuluh limfe
- Peran : pertahanan tubuh
→ proliferasi limfosit (B)
- Terdiri atas korteks dan medula
- Permukaan cembung → tempat masuk pembuluh limfe aferen
- Permukaan cekung (hilus)
 - tempat arteri masuk dan vena keluar
 - tempat keluar pembuluh limfe eferen

Mikroskopis Limfonodus

C : Capsula

T : Trabekula

Cx : Cortex

P : Paracortex

M : Medulla

MC : Medullary Cord

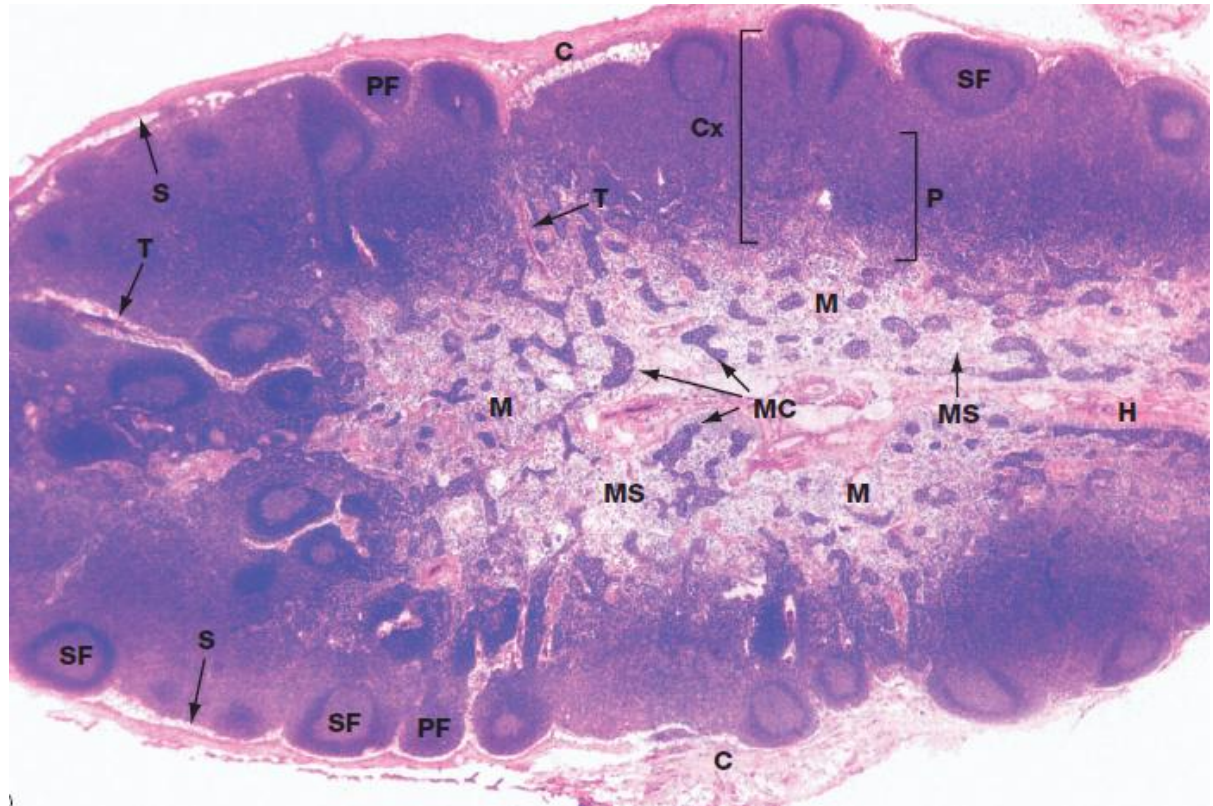
MS : medullary Sinus

SF : Secondary Folicle (aktif)

PF : Primary Folicle (inaktif)

S : Sinus subcapsular

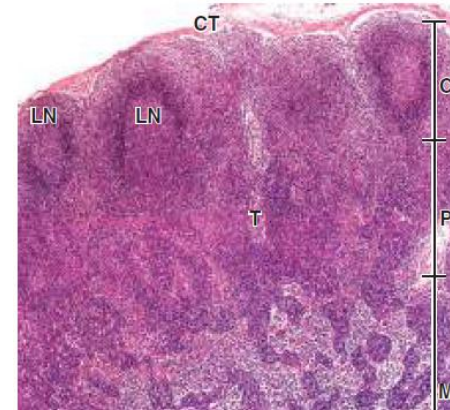
H : Hillum



Cortex Limfonodus

1. Bagian luar :

- Sinus (subkapsularis/marginalis) yg disusun oleh jaringan longgar: serabut retikuler, sel retikuler, sel makrofag



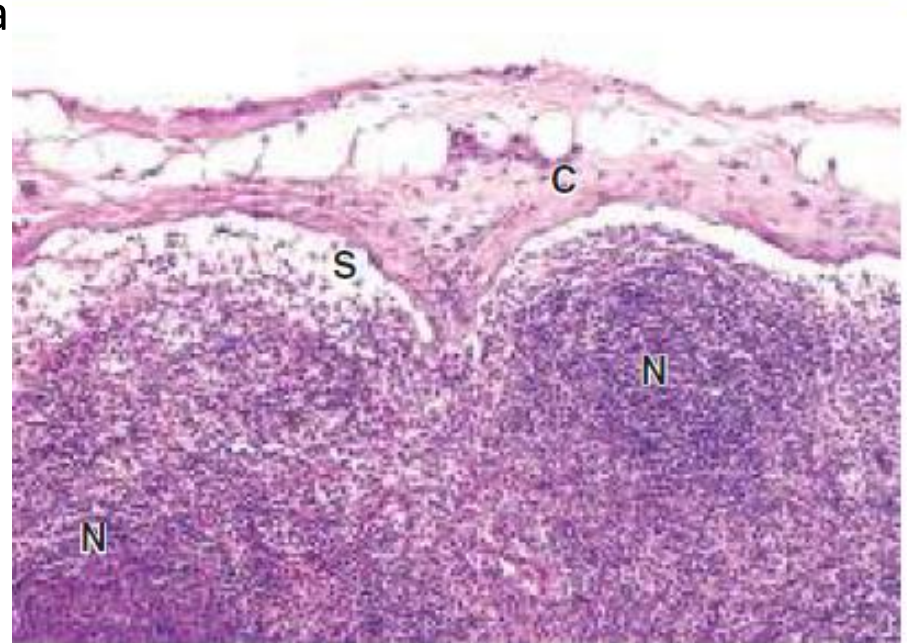
2. Bagian dalam :

- Limfonodulus (LN, penyusun utama limfosit B) & jaringan limfatika difusa

ZONA PARAKORTEKS

- batas korteks-medula
- Penyusun utama : limfosit T

FIGURE 14-17 Lymph node cortex.



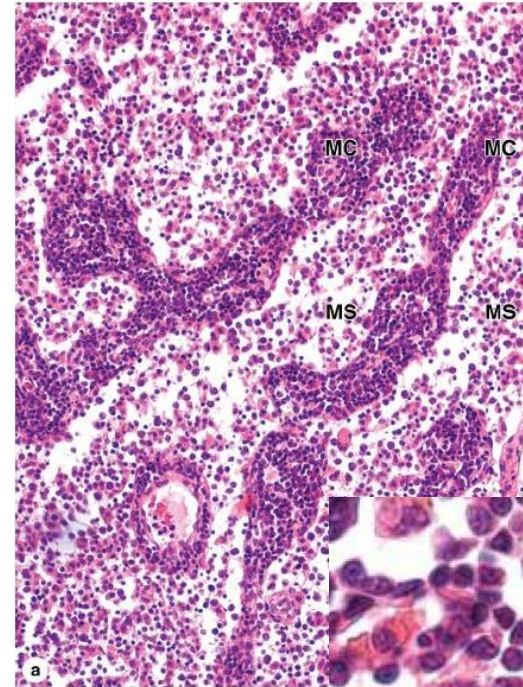
Medulla Limfonodus

1. Korda (genjel) medularis (MC) :

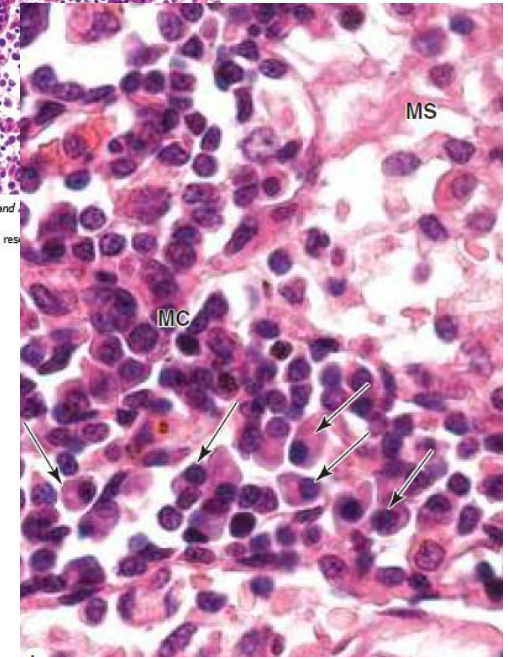
- ✓ perluasan jaringan limfoid parakortex
- ✓ sel-sel limfosit B, sel plasma, makrofag

2. Sinus medularis (MS) :

- ✓ limfe, limfosit, makrofag dan kadang2 granulosit jika terjadi infeksi
- ✓ berhubungan dengan sinus marginalis melalui sinus intermedia (trabekularis)



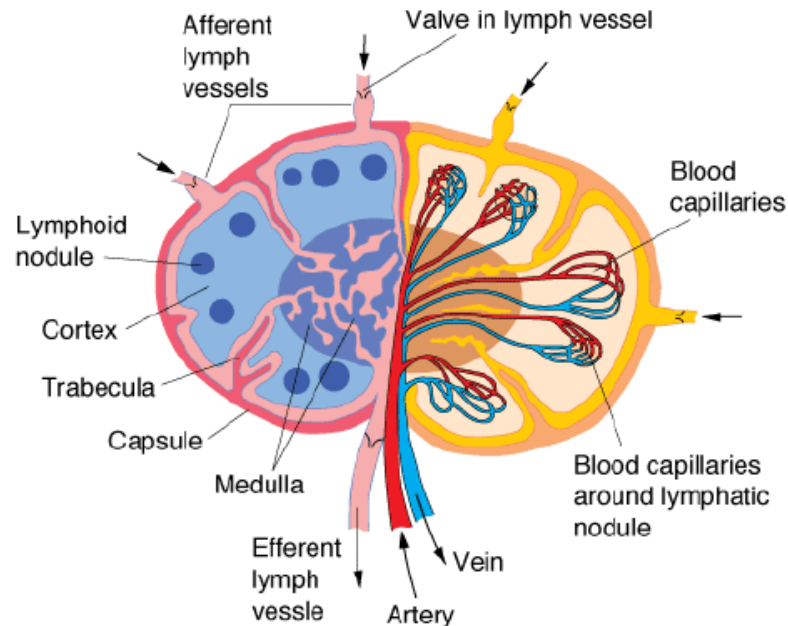
Source: Mescher AL: Junqueira's Basic Histology: Text and 12th Edition: <http://www.accessmedicine.com>
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Sirkulasi limfonodus

Aliran Limfe

Sinus marginalis → sinus trabekularis: sinus medularis & jar.
Limfatika korteks → pembuluh limfe eferen (pd hilus).



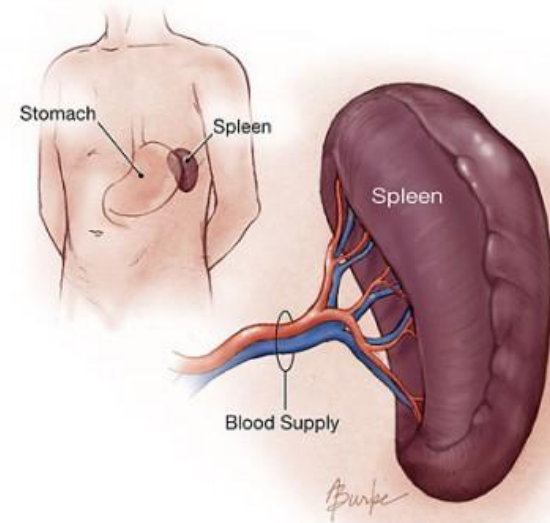
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Sirkulasi darah

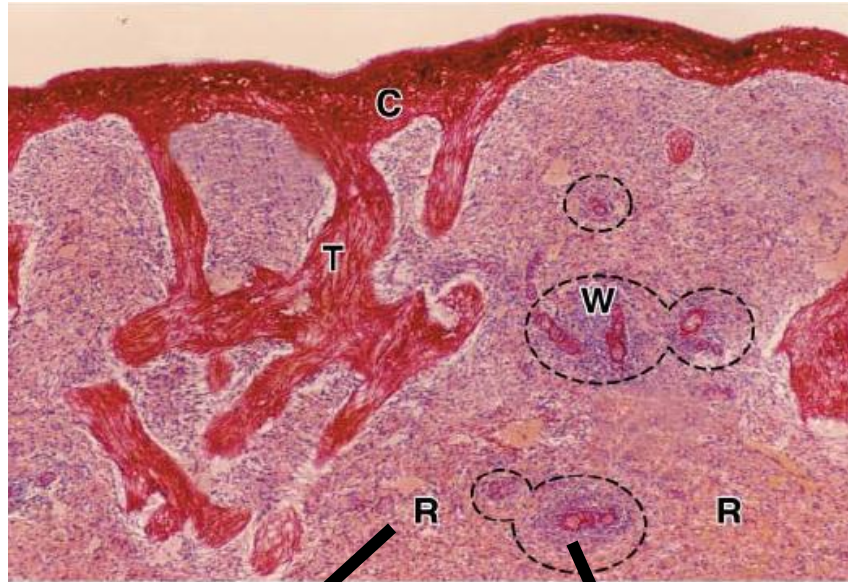
Arteri di hilum → bercabang2 di medulla → capillary networks supplying cortical follicles, paracortical zone and medullary cords → vena, keluar lewat hilum.

3. LIEN (LIMPA)

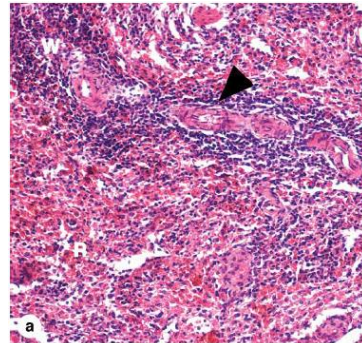
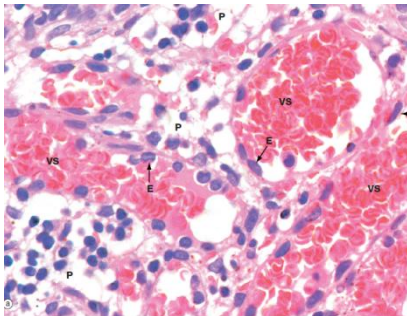
- Organ limfatika terbesar
- Fungsi :
 - ✓ produksi respon imun
 - ✓ destruksi eritrosit yang sudah tua / rusak (oleh makrofag).
 - ✓ *Recycling* iron ke sumsum tulang
 - ✓ hematopoiesis pd fetus normal dan org dewasa dg penyakit tertentu



Mikroskopis Lien



Source: Mescher AL: *Junqueira's Basic Histology: Text and Atlas*, 12th Edition: <http://www.accessmedicine.com>
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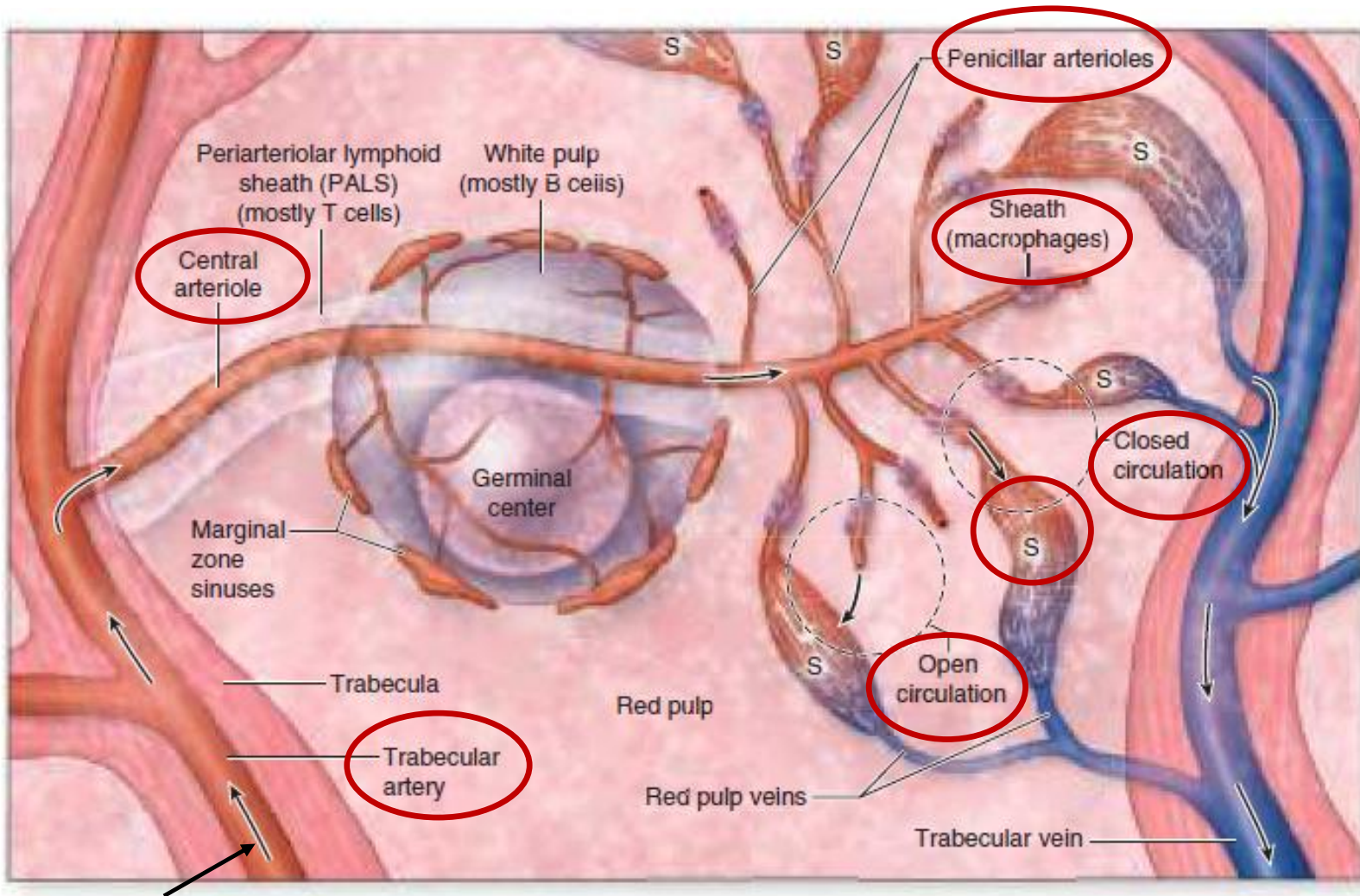
Source: Mescher AL: *Junqueira's Basic Histology: Text and Atlas*, 12th Edition: <http://www.accessmedicine.com>
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Kapsula (C) : jar. Ikat padat yg membentuk trabekula (T)

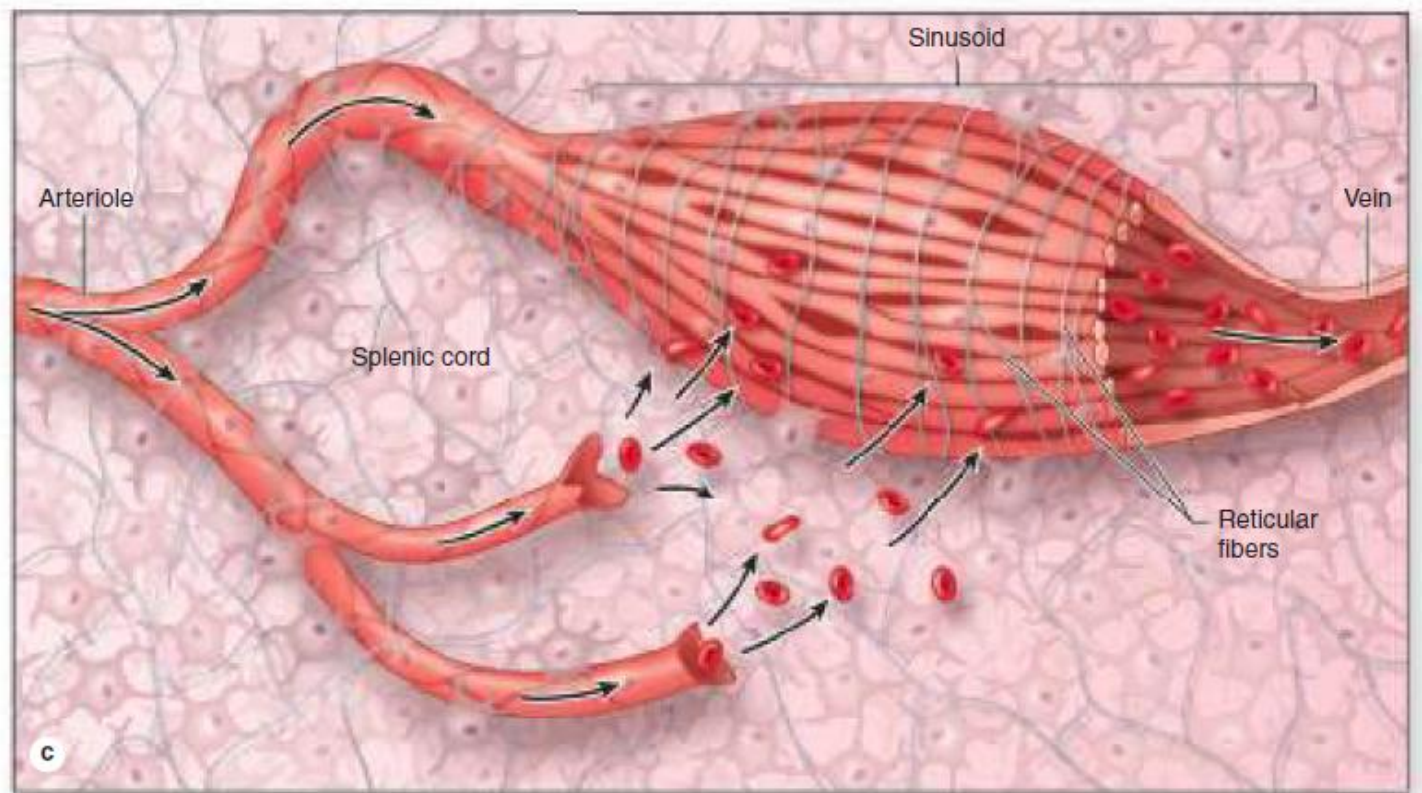
Parenkim/pulpa lien :

1. Pulpa putih (W)
 - ✓ jar.limfoid
 - ✓ di sekitar arteriol sentralis, membentuk *periarteriolar lymphoid sheath (PALS)*
 - ✓ kondisi segar tampak sbg bintik2 putih.
2. Pulpa merah (R)
 - ✓ korda (cords of billroth) dan sinusoid
 - ✓ berisi semua tipe sel darah
 - ✓ kondisi segar tampak merah tua

VASKULARISASI LIEN



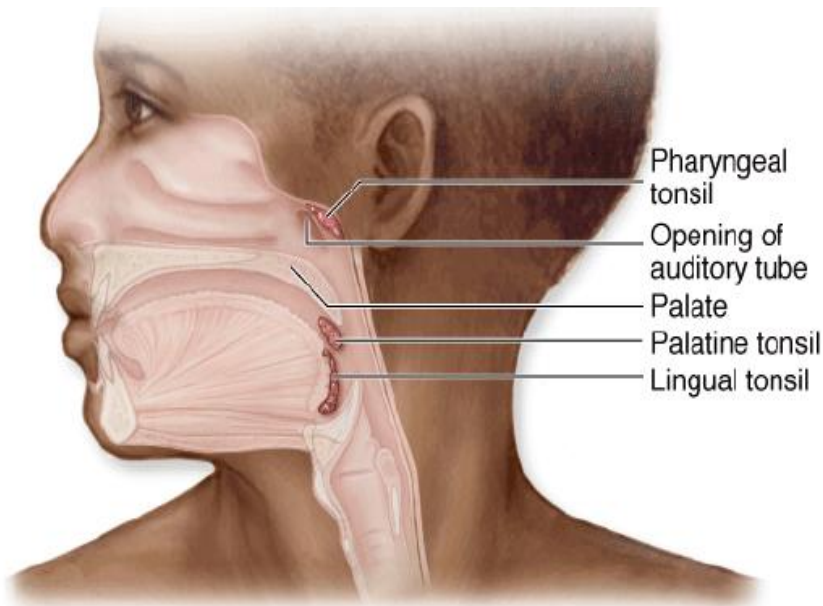
Arteri lienalis



4. Mucosa-associated Lymphoid Tissue (MALT)

- Lumen saluran pencernaan, pernafasan dan genitourinari berhubungan dg dunia luar → invasi patogen.
- Utk melindunginya → *mucosa-associated lymphoid tissue (MALT)* ex : tonsil, payers pacht.
- Berdasarkan tempatnya :
 - Sal. Cerna → *gut-associated lymphoid tissue (GALT)*
 - Bronkus → *bronchial-associated lymphoid tissue (BALT)*

TONSIL

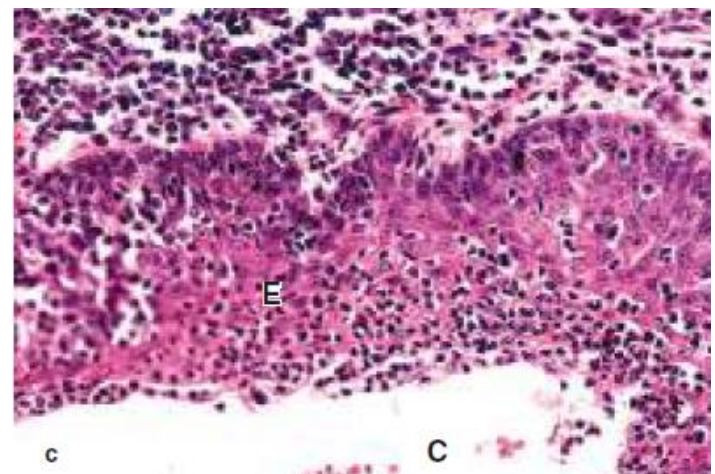
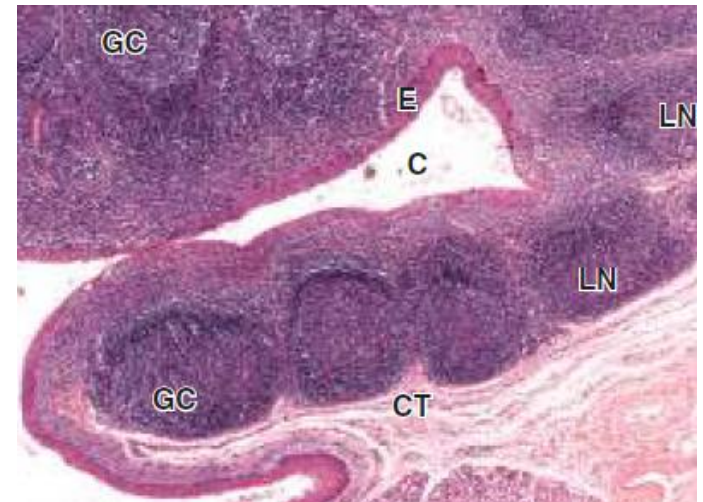


Source: Mescher AL: *Junqueira's Basic Histology: Text and Atlas*, 12th Edition: <http://www.accessmedicine.com>

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- Tonsil palatina & lingualis → *stratified squamous epithelium*
- Tonsil pharyngeal → pseudostratified columnar epithelium, stratified epithelium

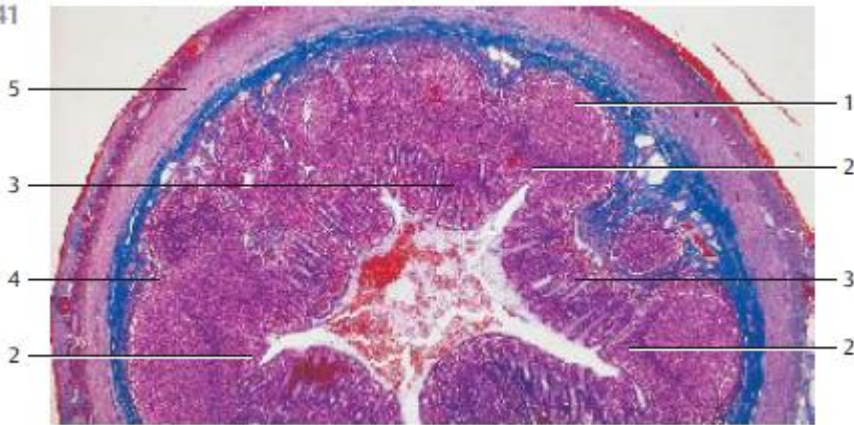
Tonsil



**E = epithel; C = Crypte; F = Folikel
Limfoid, Cap = Colagenous Capsule,
GC = Germinal Centre**

Gut-associated Lymphoid Tissue (GALT)

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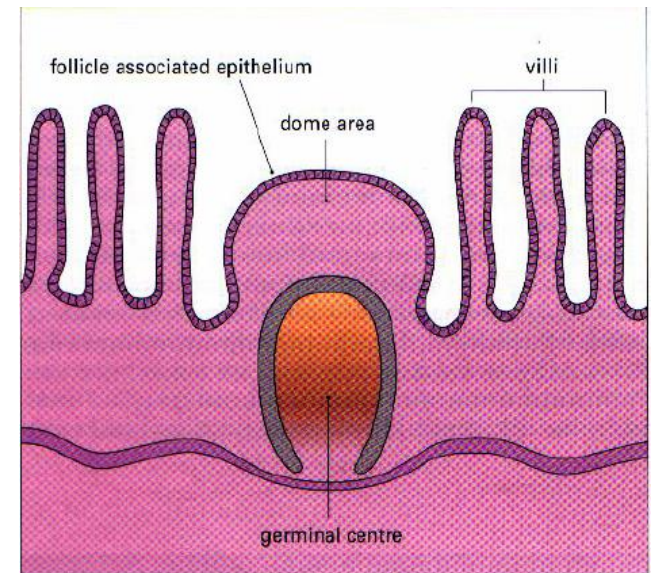


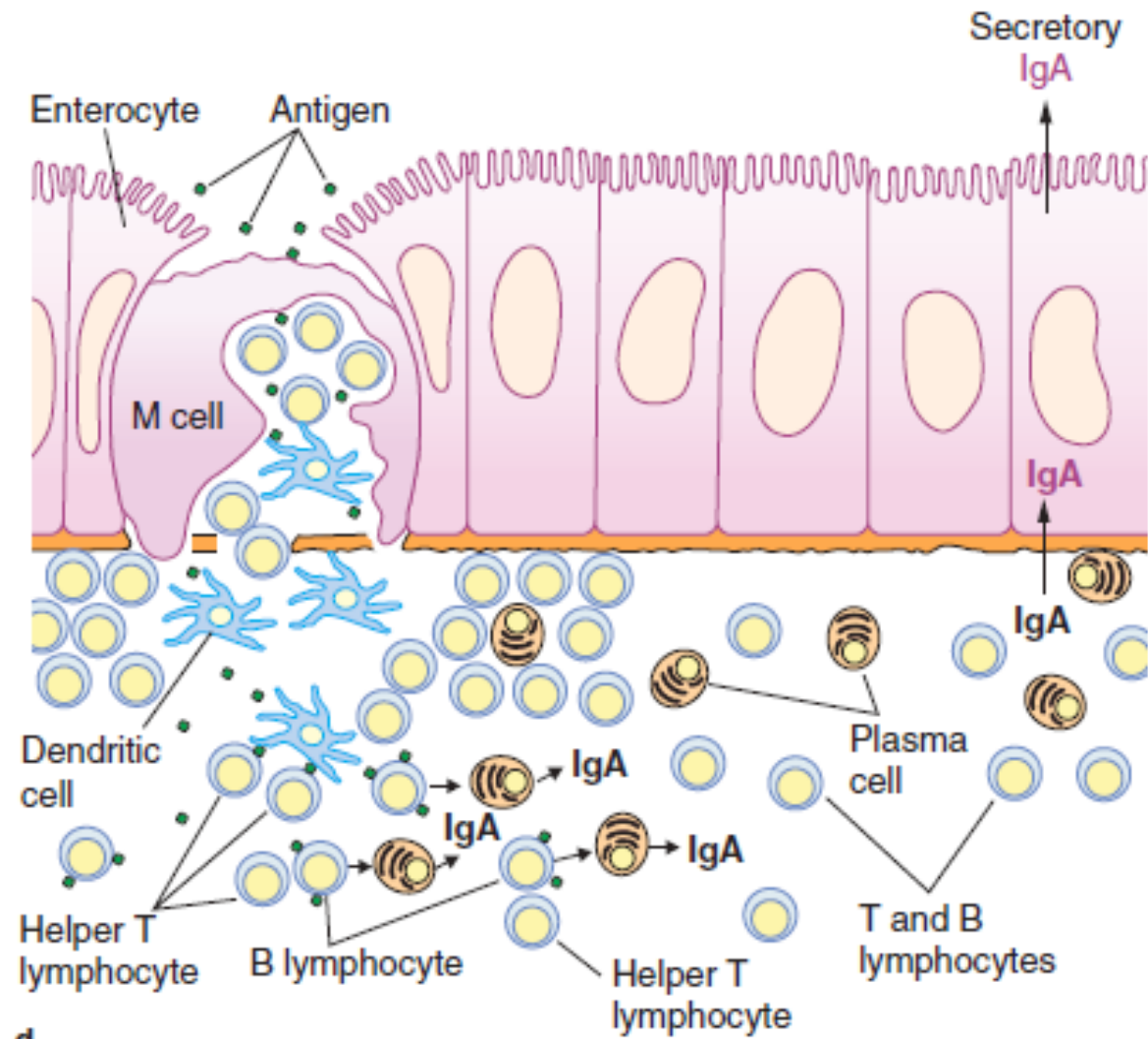
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A. Appendix. 1. Follikel limfoid. 2. Dome area. 3. membran mukosa. 4. interfollicular region. 5. Tunika muskularis.

B. Ileum. N= limphoidnodule

- Peyer's patches (groups of lymphoid follicles) bulge dome-like into intestinal lumen → dome area.
- Dome area dilapisi epitel (*Follicle-associated epithelium* (FAE)) dg ciri : tdk ada mikrofili, sel goblet dan kripte, namun mpy microfold (*microfolds cell*, sel M)





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TERIMA KASIH

